INCH-POUND

MIL-M-28787/437A
7 September 1994
SUPERSEDING
MIL-M-28787/437
6 August 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MAINTENANCE PROCESSOR SERIAL INTERFACE MODULE, KEY CODE HRAA

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic Maintenance Processor Serial Interface module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Code 603A, Bldg. 2917, NAVSURFWARCENDIV, 300 Highway 361, Crane, IN 47522-5001, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 24.983 failures per million operating hours for class I or 85.947 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be HRAA.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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* TABLE I. Module electrical requirements.

	Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
					Min	Max	Min	Max	Min	Max	
*	A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
		End-of-life	II	S	1	INF	1	INF	1	INF	MOhm
*	A2	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
		End-of-life	II	X	1	INF	1	INF	1	INF	MOhm
	B1-B8	Continuity	III	X	0	1	0	1	0	1	Ohm
	C	Vcc stress	IV	Q	----------Module shall pass all remaining tests----------						
	D1	Ibd	V	S	0 0	1 0	0 0	1 0	0.0	1 0	mA
	D2	Ibd	V	S	-0 06	1.1	-0 06	1 1	-0 06	1.1	mA
	D3,D4	Ibd	V	S	0 0	0.1	0 0	0 1	0.0	0 1	mA
	D5	Ibd	V	S	0.0	1 0	0 0	1 0	0.0	1.0	mA
	D6-D22	Ibd	V	S	0 0	0 1	0 0	0 1	0.0	0.1	mA
	D23	Ibd	V	S	0 0	2 1	0 0	2 1	0.0	2.1	mA
	D24	Ibd	V	S	-0 06	1 1	-0 06	1 1	-0.06	1.1	mA
	D25	Ibd	V	S	0 0	2 3	0 0	2.3	0 0	2 3	mA
	D26	Ibd	V	S	0.0	0 1	0 0	0 1	0.0	0 1	mA
	E1-E8	Ios	VI	S	-150 0	-6 0	-150 0	-6.0	-150.0	-6 0	mA
	E9-E12	Ios	VI	S	-150.0	-20 0	-150 0	-20 0	-150 0	-20 0	mA
	E13	Ios	VI	S	-85.0	-15 0	-85.0	-15.0	-85.0	-15.0	mA
	E14-E19	Ios	VI	S	-76 4	-23 6	-76 4	-23.6	-76.4	-23.6	mA
*	E20	Ios	VI	S	-2.92	-2 59	-2 92	-2.59	-2 92	-2 59	mA
*	E21-E28	Ios	VI	S	-225 0	-40 0	-225 0	-40.0	-225.0	-40.0	mA
*	E29-E32	Ios	VI	S	-150 0	-20 0	-150 0	-20 0	-150.0	-20 0	mA
*	E33,E34	Ios	VI	S	-76 4	-23 6	-76 4	-23.6	-76.4	-23 6	mA
*	E35-E42	Ios	VI	S	-150 0	-30 0	-150 0	-30 0	-150.0	-30.0	mA
	F1-F8	Iih	VII	X	-2 9	2 4	-2 9	2 4	-2 9	2.4	mA
	F9	Iih	VII	X	-0 25	0 025	-0 25	0.025	-0 25	0 025	mA
	F10	Iih	VII	X	-0 5	0 05	-0.5	0 05	-0 5	0.05	mA
	F11	Iih	VII	X	-2.95	-1 41	-2 95	-1.41	-2 95	-1 41	mA
	F12	Iih	VII	X	-0 1	0 02	-0 1	0 02	-0 1	0 02	mA
	F13	Iih	VII	X	-0 36	0 02	-0 36	0.02	-0 36	0.02	mA

- See footnotes at end of table

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• TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F14-F17	IIh	VII	X	-2 9	2.4	-2.9	2.4	-2 9	2.4	mA
F18	IIh	VII	X	-0.25	0 025	-0.25	0 025	-0.25	0.025	mA
F19	IIh	VII	X	-0 36	0.02	-0.36	0 02	-0 36	0.02	mA
F20-F27	IIh	VII	X	-2.8	2 3	-2 8	2.3	-2 8	2.3	mA
F28-F35	IIh	VII	X	-0 2	0 02	-0.2	0 02	-0 2	0.02	mA
F36	IIh	VII	X	-1 15	0 095	-1 15	0.095	-1 15	0 095	mA
F37-F44	IIh	VII	X	-0 2	0 02	-0 2	0 02	-0 2	0 02	mA
F45	IIh	VII	X	-0 7	0 07	-0 7	0 07	-0 7	0 07	mA
F46-F61	IIh	VII	X	-2 8	2 3	-2.8	2.3	-2 8	2 3	mA
F62	IIh	VII	X	-2 7	-1 4	-2 7	-1 4	-2 7	-1 4	mA
F63	IIh	VII	X	-1 7	0 11	-1 7	0.11	-1 7	0 11	mA
F64	IIh	VII	X	-0 1	0 02	-0 1	0 02	-0 1	0 02	mA
G1-G8	IIl	VIII	X	-2.9	2 4	-2 9	2.4	-2 9	2 4	mA
G9	IIl	VIII	X	-0.25	0 025	-0.25	0.025	-0 25	0 025	mA
G10	IIl	VIII	X	-0 5	0 05	-0 5	0.05	-0.5	0 05	mA
G11	IIl	VIII	X	-2 95	-1 41	-2 95	-1 41	-2 95	-1 41	mA
G12	IIl	VIII	X	-0 1	0 02	-0.1	0 02	-0 1	0 02	mA
G13	IIl	VIII	X	-0 36	0 02	-0 36	0 02	-0 36	0 02	mA
G14-G17	IIl	VIII	X	-2 9	2 4	-2 9	2 4	-2 9	2 4	mA
G18	IIl	VIII	X	-0 25	0 025	-0.25	0.025	-0 25	0.025	mA
G19	IIl	VIII	X	-0 36	0 02	-0 36	0.02	-0 36	0 02	mA
G20-G27	IIl	VIII	X	-2 8	2 3	-2 8	2 3	-2 8	2.3	mA
G28-G35	IIl	VIII	X	-0 2	0 02	-0 2	0.02	-0 2	0 02	mA
G36	IIl	VIII	X	-1 15	0 095	-1 15	0 095	-1.15	0.095	mA
G37-G44	IIl	VIII	X	-0 2	0.02	-0.2	0.02	-0 2	0 02	mA
G45	IIl	VIII	X	-0.7	0 07	-0 7	0 07	-0.7	0.07	mA
G46-G61	IIl	VIII	X	-2.8	2 3	-2 8	2 3	-2 8	2 3	mA
G62	IIl	VIII	X	-2 7	-1 4	-2 7	-1 4	-2 7	-1 4	mA
G63	IIl	VIII	X	-1 7	0 11	-1 7	0 11	-1 7	0.11	mA
G64	IIl	VIII	X	-0 1	0 02	-0 1	0 02	-0 1	0 02	mA
H1-H8	Voh	IX	X	2 0	6 0	2 0	6 0	2 0	6 0	V
H9-H12	Voh	IX	X	3 6	6 0	3.6	6.0	3.6	6 0	V
H13-H27	Voh	IX	X	2 4	4 5	2 4	4 5	2.4	4 5	V
H28-H31	Voh	IX	X	3 6	6 0	3.6	6.0	3.6	6.0	V

- See footnotes at end of table

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* TABLE I Module electrical requirements - Continued.

	Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
					Min	Max	Min	Max	Min	Max	
	H32-H41	Voh	IX	X	2.4	4 5	2 4	4 5	2 4	4.5	V
	J1-J8	Vol	X	X	-6 0	-2 0	-6 0	-2 0	-6 0	-2 0	V
	J9-J12	Vol	X	X	-6.0	-3 6	-6 0	-3 6	-6 0	-3.6	V
	J13-J28	Vol	X	X	0 0	0 5	0 0	0 5	0 0	0 5	V
	J29-J32	Vol	X	X	-6 0	-3 6	-6 0	-3 6	-6 0	-3 6	V
	J33-J44	Vol	X	X	0 0	0 5	0 0	0 5	0 0	0 5	V
*	K1-K16	Iozl	XI	X	-0 02	0 02	-0 02	0 02	-0 02	0 02	mA
*	K17,K18	Iozh	XI	X	0 0	0 01	0 0	0.01	0 0	0 01	mA
*	K19-K34	Iozh	XI	X	-0 02	0 02	-0 02	0 02	-0.02	0 02	mA
	L1-L24	Vic	XII	S	-1 5	0 0	-1 5	0 0	-1 5	0 0	V
	M1	PSFC	XIII	S	5 9	10 9	5 9	10.9	5.9	10 9	µF
	M2	PSFC	XIII	S	0 8	1 5	0 8	1.5	0.8	1 5	µF
	N1	Icc	XIV	X	680	1633	680	1633	680	1633	mA
	N2	Icc	XIV	X	-22	-7	-22	-7	-22	-7	mA
	P1,P2	Functional	XV	X	See table XV						
	Q1,Q2	Dynamic	XVI	S	See table XVI						
	Q3,Q4	Dynamic	XVI	Q	See table XVI						
	R1	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
	R2	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
	R3-R6	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
	R7-R27	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
	R28	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
	R29,R30	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
	R31	Ci/o	XVIII	Q	2	125	2	125	2	125	pF
	R32-R47	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
	R48	Ci/o	XVIII	Q	4	200	4	200	4	200	pF
	R49	Ci/o	XVIII	Q	8	200	8	200	8	200	pF
	R50	Ci/o	XVIII	Q	2	75	2	75	2	75	pF

- See footnotes at end of table

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* TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R51-R59	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R60-R63	Ci/o	XVIII	Q	6	100	6	100	6	100	pF
R64	Ci/o	XVIII	Q	8	175	8	175	8	175	pF
R65-R68	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
T1,T2	PSTA	XIX	Q	0	400	0	400	0	400	mV

1/ "X" parameters shall be 100 percent tested during the module acceptance test

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C

3/ "Q" parameters shall be tested during qualification testing only

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/437-1 for class I and M28787/437-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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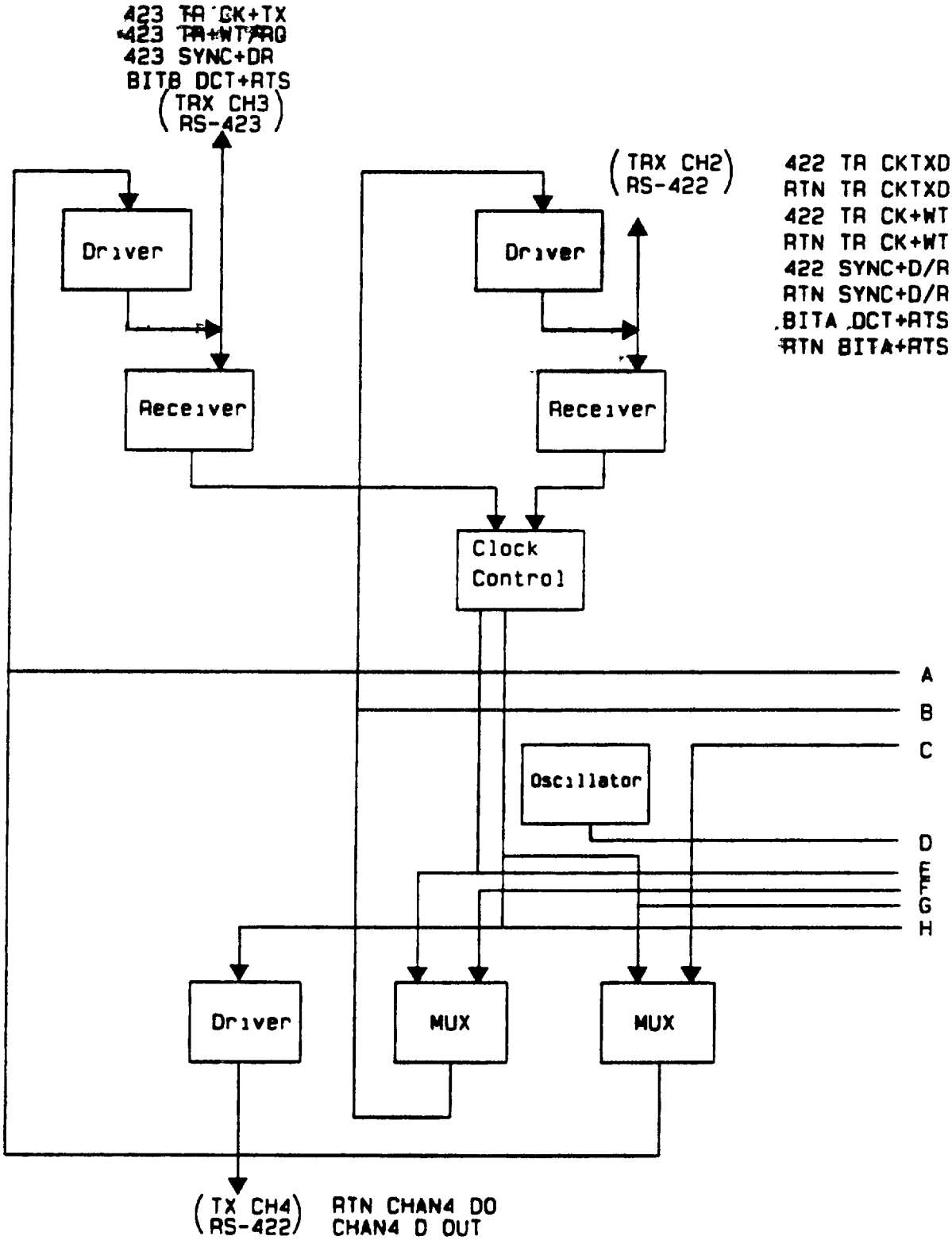


FIGURE 1. HRAA functional block diagram.

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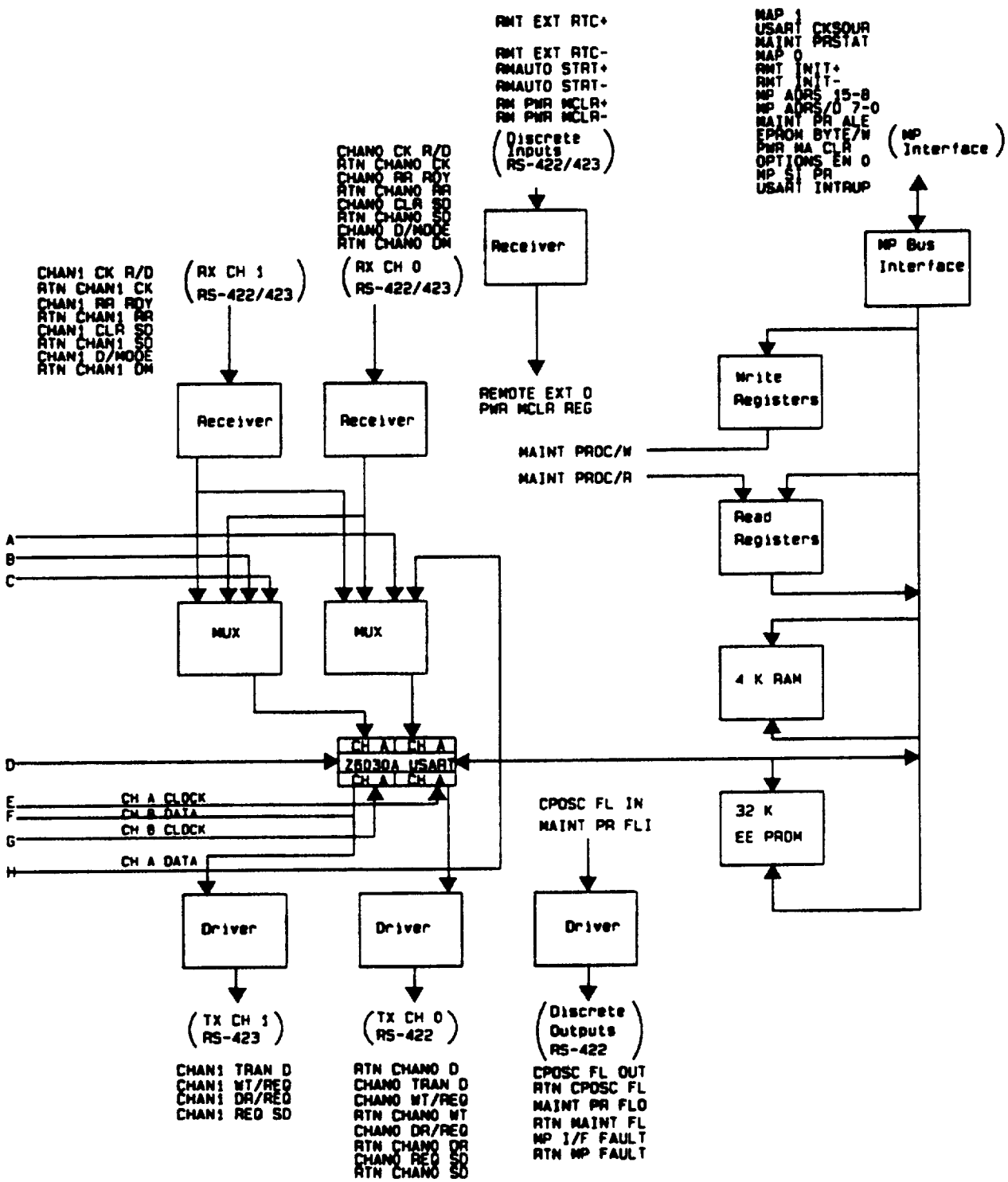
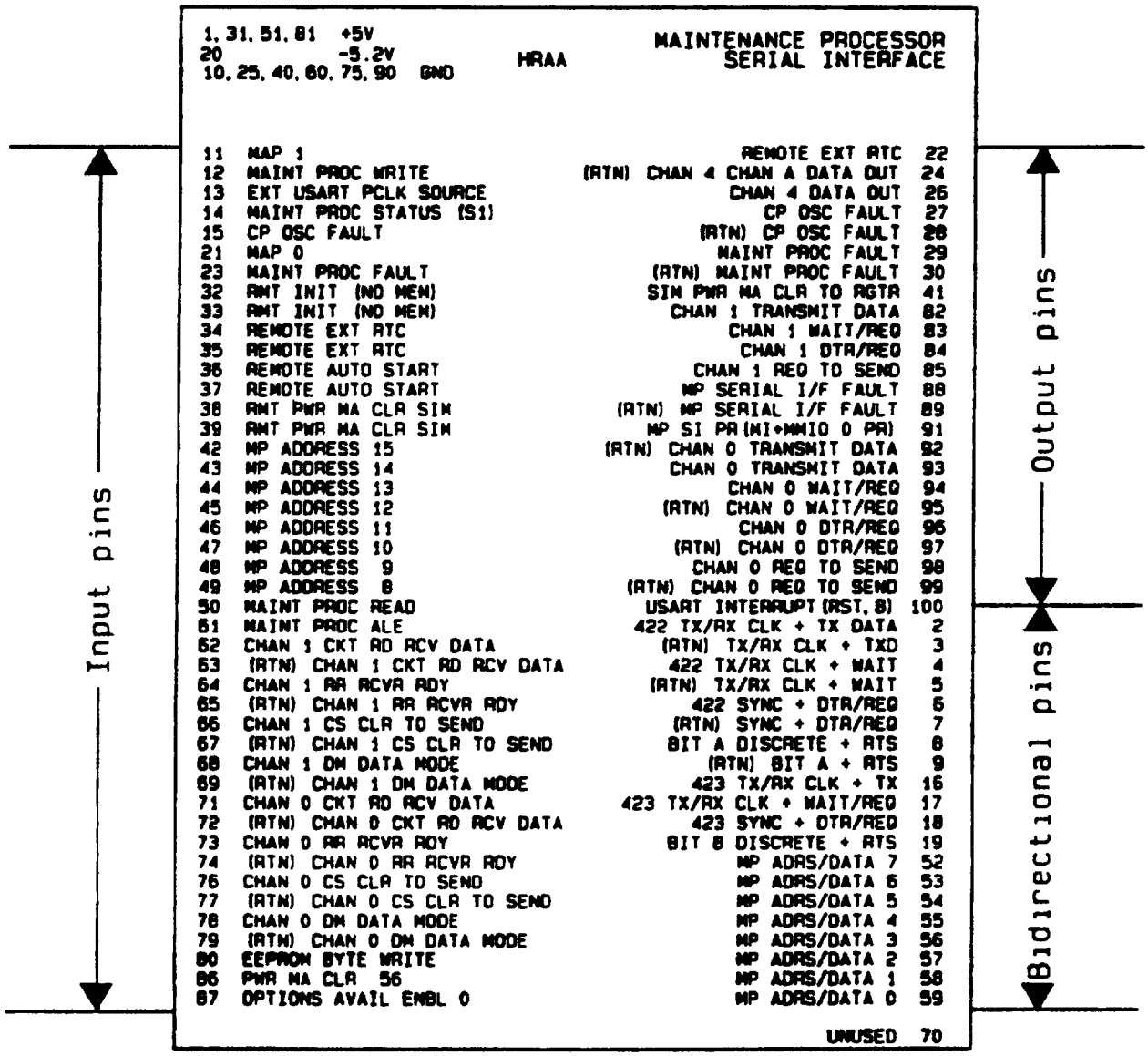


FIGURE 1. HRAA functional block diagram - Continued.

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* FIGURE 2. HRAA pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

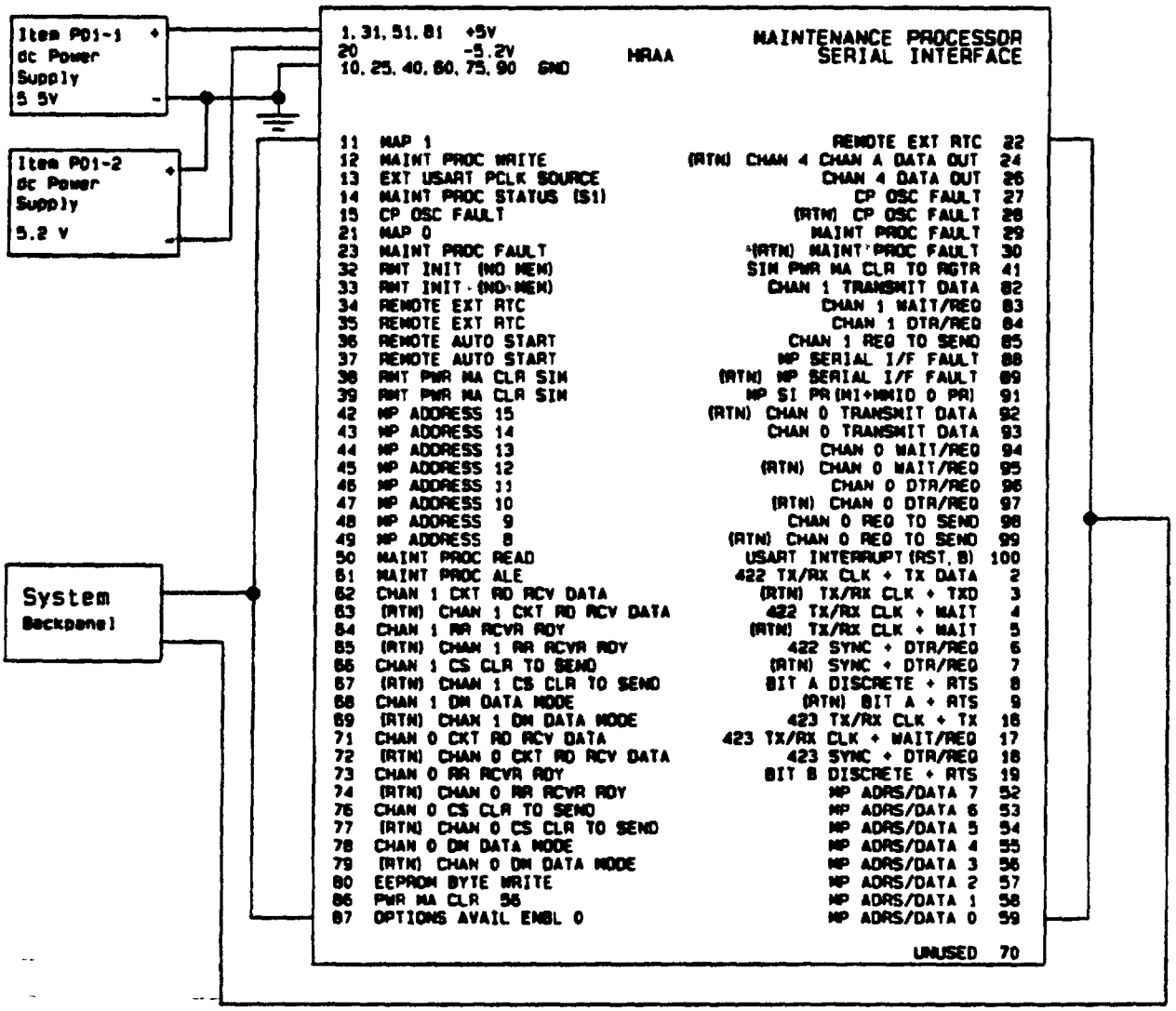
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

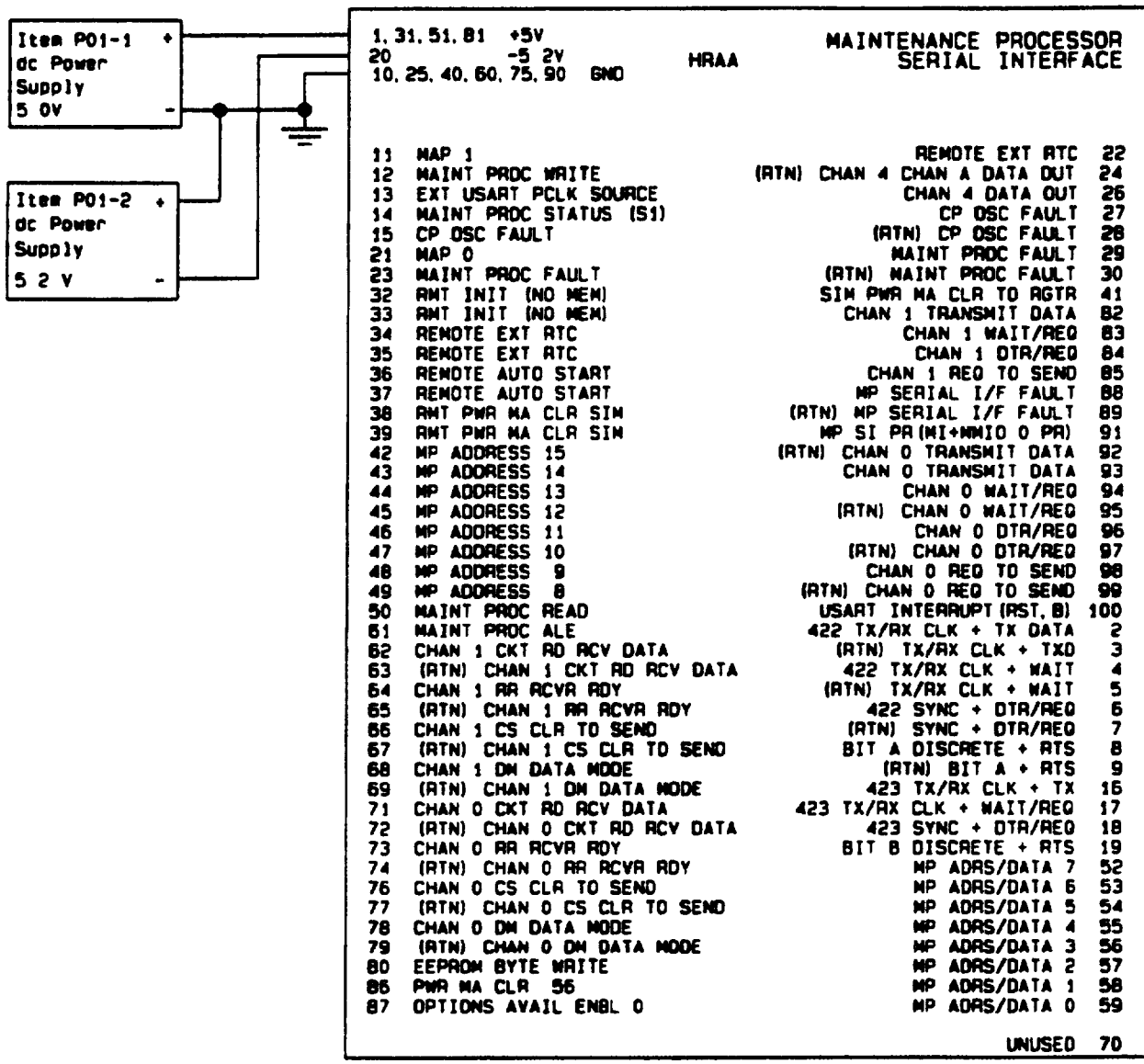
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NOTE:
1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

*FIGURE 3. HRAA operating temperature and component temperature test circuit.

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* FIGURE 4. HRAA vibration test circuit.

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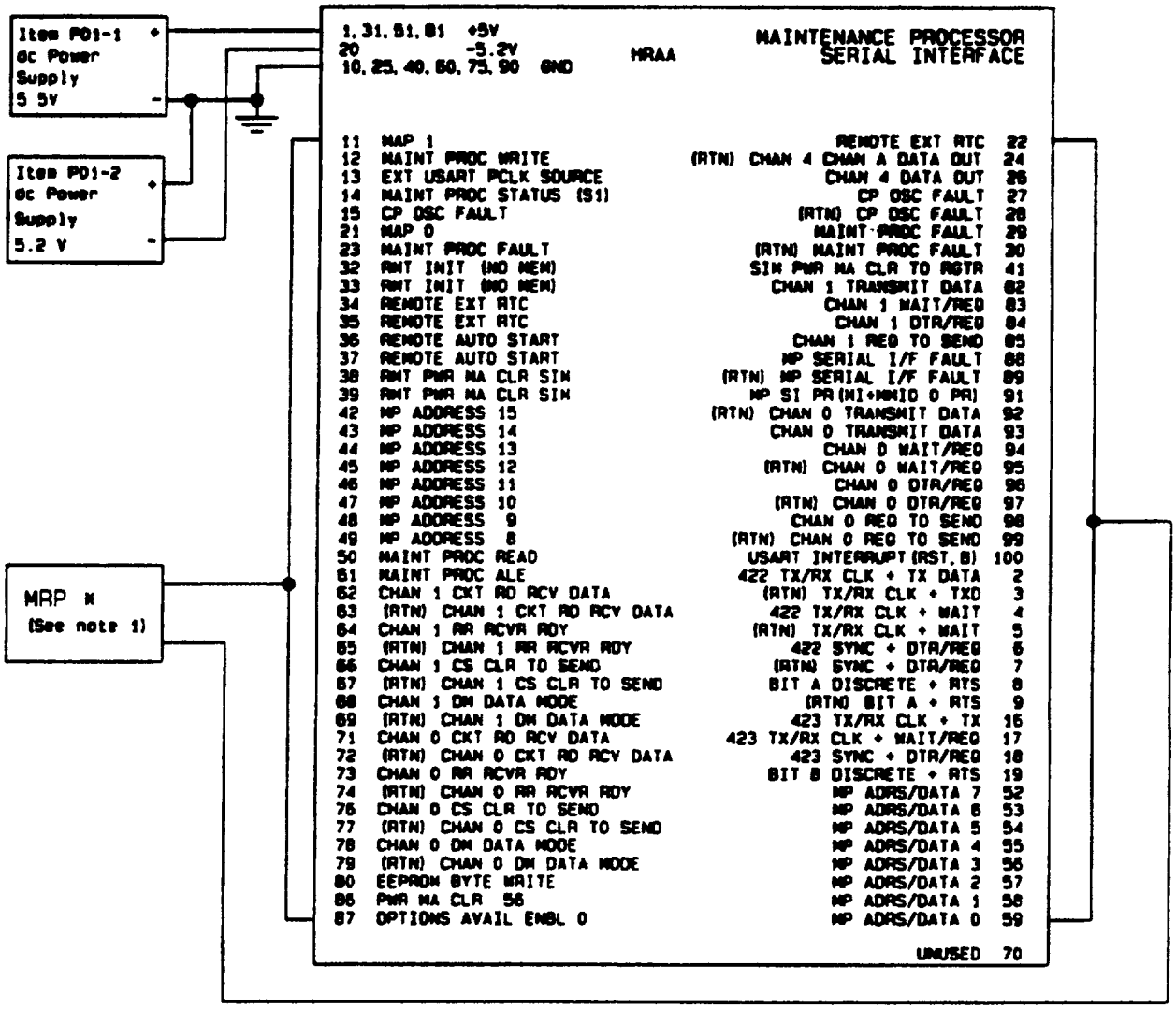
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

* FIGURE 5. HRAA life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: (Vcc) 4.5 Vdc to 5.5 Vdc, (V2) -5.72 Vdc to -4.68 Vdc. Module power dissipation: 9.107 W (max); 6.1 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
422 TR CKTXD	NA	0.4	2.4	-2.9	2.4	NA	75
RTN TR CKTXD	NA	0.4	2.4	-2.9	2.4	NA	75
422 TR CK+WT	NA	0.4	2.4	-2.9	2.4	NA	75
RTN TR CK+WT	NA	0.4	2.4	-2.9	2.4	NA	75
422 SYNC+D/R	NA	0.4	2.4	-2.9	2.4	NA	75
RTN SYNC+D/R	NA	0.4	2.4	-2.9	2.4	NA	75
BITA DCT+RTS	NA	0.4	2.4	-2.9	2.4	NA	75
RTN BITA+RTS	NA	0.4	2.4	-2.9	2.4	NA	75
MAP 1	-1.5	0.4	2.4	-0.25	0.025	1.0	50
MAINT PROC/W	NA	0.4	2.4	-0.5	0.05	NA	75
USART CKSOUR	NA	0.4	2.4	-2.95	-1.41	1.1	75
MAINT PRSTAT	-1.5	0.4	2.4	-0.1	0.02	0.1	75
CPOSC FL IN	-1.5	0.4	2.4	-0.36	0.02	0.1	75
423 TR CK+TX	NA	0.4	2.4	-2.9	2.4	NA	100
423 TR+WT/RQ	NA	0.4	2.4	-2.9	2.4	NA	100
423 SYNC+D/R	NA	0.4	2.4	-2.9	2.4	NA	100
BITB DCT+RTS	NA	0.4	2.4	-2.9	2.4	NA	100
MAP 0	1.5	0.4	2.4	-0.25	0.025	1.0	75
MAINT PR FLI	1.5	0.4	2.4	-0.36	0.02	0.1	75
RMT INIT +	NA	0.4	2.4	-2.8	2.3	NA	50
RMT INIT -	NA	0.4	2.4	-2.8	2.3	NA	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
RMT EXT RTC +	NA	0.4	2.4	-2.8	2.3	NA	50
RMT EXT RTC -	NA	0.4	2.4	-2.8	2.3	NA	50
RMAUTO STRT +	NA	0.4	2.4	-2.8	2.3	NA	50
RMAUTO STRT -	NA	0.4	2.4	-2.8	2.3	NA	50
RM PWR MCLR +	NA	0.4	2.4	-2.8	2.3	NA	50
RM PWR MCLR -	NA	0.4	2.4	-2.8	2.3	NA	50
MP ADRS 15	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MP ADRS 14	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MP ADRS 13	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MP ADRS 12	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MP ADRS 11	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MP ADRS 10	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MP ADRS 9	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MP ADRS 8	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MAINT PROC/R	NA	0.4	2.4	-1.15	0.095	NA	175
MP ADRS/D 7	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MP ADRS/D 6	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MP ADRS/D 5	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MP ADRS/D 4	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MP ADRS/D 3	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MP ADRS/D 2	-1.5	0.4	2.4	-0.2	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MP ADRS/D 1	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MP ADRS/D 0	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MAINT PR ALE	-1.5	0.4	2.4	-0.7	0.07	2.1	125
CHAN1 CK R/D	NA	0.4	2.4	-2.8	2.3	NA	50
RTN CHAN1 CK	NA	0.4	2.4	-2.8	2.3	NA	50
CHAN1 RR RDY	NA	0.4	2.4	-2.8	2.3	NA	50
RTN CHAN1 RR	NA	0.4	2.4	-2.8	2.3	NA	50
CHAN1 CLR SD	NA	0.4	2.4	-2.8	2.3	NA	50
RTN CHAN1 SD	NA	0.4	2.4	-2.8	2.3	NA	50
CHAN1 D/MODE	NA	0.4	2.4	-2.8	2.3	NA	50
RTN CHAN1 DM	NA	0.4	2.4	-2.8	2.3	NA	50
CHAN0 CK R/D	NA	0.4	2.4	-2.8	2.3	NA	50
RTN CHAN0 CK	NA	0.4	2.4	-2.8	2.3	NA	50
CHAN0 RR RDY	NA	0.4	2.4	-2.8	2.3	NA	50
RTN CHAN0 RR	NA	0.4	2.4	-2.8	2.3	NA	50
CHAN0 CLR SD	NA	0.4	2.4	-2.8	2.3	NA	50
RTN CHAN0 SD	NA	0.4	2.4	-2.8	2.3	NA	50
CHAN0 D/MODE	NA	0.4	2.4	-2.85	2.3	NA	50
RTN CHAN0 DM	NA	0.4	2.4	-2.8	2.3	NA	50
EEPROM BYTE/W	NA	0.4	2.4	-2.7	-1.4	1.1	200
PWR MA CLR	-1.5	0.4	2.4	-1.7	0.11	2.3	200
OPTIONS EN 0	-1.5	0.4	2.4	-0.1	0.02	0.1	75

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* 6.3.2 Output parameters.

	Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
	422 TR CKTXD	2.0	-2.0	-0.16	1.6	NA	NA	-6.0	75
	RTN TR CKTXD	2.0	-2.0	-0.16	1.6	NA	NA	-6.0	75
	422 TR CK+WT	2.0	-2.0	-0.16	1.6	NA	NA	-6.0	75
	RTN TR CK+WT	2.0	-2.0	-0.16	1.6	NA	NA	-6.0	75
	422 SYNC+D/R	2.0	-2.0	-0.16	1.6	NA	NA	-6.0	75
	RTN SYNC+D/R	2.0	-2.0	-0.16	1.6	NA	NA	-6.0	75
	BITA DCT+RTS	2.0	-2.0	-0.16	1.6	NA	NA	-6.0	75
	RTN BITA+RTS	2.0	-2.0	-0.16	1.6	NA	NA	-6.0	75
	423 TR CK+TX	3.6	-3.6	-0.16	1.6	NA	NA	-20.0	100
	423 TR+WT/RQ	3.6	-3.6	-0.16	1.6	NA	NA	-20.0	100
	423 SYNC+D/R	3.6	-3.6	-0.16	1.6	NA	NA	-20.0	100
	BITB DCT+RTS	3.6	-3.6	-0.16	1.6	NA	NA	-20.0	100
	REMOTE EXT O	2.4	0.5	-0.16	1.6	NA	NA	-15.0	NA
*	RTN CHAN4 DO	2.4	0.5	-0.1	1.0	0.02	0.02	-23.6	NA
*	CHAN4 D OUT	2.4	0.5	-0.1	1.0	0.02	0.02	-23.6	NA
*	CPOSC FL OUT	2.4	0.5	-0.1	1.0	0.02	0.02	-23.6	NA
*	RTN CPOSC FL	2.4	0.5	-0.1	1.0	0.02	0.02	-23.6	NA
*	MAINT PR FLO	2.4	0.5	-0.1	1.0	0.02	0.02	-23.6	NA
*	RTN MAINT FL	2.4	0.5	-0.1	1.0	0.02	0.02	-23.6	NA
*	PWR MCLR REG	NA	0.5	NA	1.6	NA	NA	-2.59	NA
	MP ADRS/D 7	2.4	0.5	-0.16	1.6	NA	NA	-40.0	50

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* 6.3.2 Output parameters - Continued.

	Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
	MP ADRS/D 6	2.4	0.5	-0.16	1.6	NA	NA	-40.0	50
	MP ADRS/D 5	2.4	0.5	-0.16	1.6	NA	NA	-40.0	50
	MP ADRS/D 4	2.4	0.5	-0.16	1.6	NA	NA	-40.0	50
	MP ADRS/D 3	2.4	0.5	-0.16	1.6	NA	NA	-40.0	50
	MP ADRS/D 2	2.4	0.5	-0.16	1.6	NA	NA	-40.0	75
	MP ADRS/D 1	2.4	0.5	-0.16	1.6	NA	NA	-40.0	50
	MP ADRS/D 0	2.4	0.5	-0.16	1.6	NA	NA	-40.0	50
	CHAN1 TRAN D	3.6	-3.6	-0.16	1.6	NA	NA	-20.0	75
	CHAN1 WT/REQ	3.6	-3.6	-0.16	1.6	NA	NA	-20.0	75
	CHAN1 DR/REQ	3.6	-3.6	-0.16	1.6	NA	NA	-20.0	75
	CHAN1 REQ SD	3.6	-3.6	-0.16	1.6	NA	NA	-20.0	75
*	MP I/F FAULT	2.4	0.5	-0.1	1.0	0.02	0.02	-23.6	NA
*	RTN MP FAULT	2.4	0.5	-0.1	1.0	0.02	0.02	-23.6	NA
	MP SI PR	NA	0.5	NA	1.6	0.01	NA	-30.0	NA
*	RTN CHAN0 D	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA
*	CHAN0 TRAN D	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA
*	CHAN0 WT/REQ	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA
*	RTN CHAN0 WT	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA
*	CHAN0 DR/REQ	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA
*	RTN CHAN0 DR	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA
*	CHAN0 REQ SD	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA

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* 6.3.2 Output parameters - Continued.

	Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
*	RTN CHAN0 SD	2.4	0.5	-0.16	1.6	0.02	0.02	-30.0	NA
	USART INTRUP	NA	0.5	NA	1.6	0.01	NA	NA	NA

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	OPTIONS EN 0 to MP SI PR	-	-	-	31	31	31	nS	11a, 11b
2		CPOSC FL IN to CPOSC FL OUT, RTN CPOSC FL	-	-	-	77	77	77		
3		MAINT PR FLI to MAINT PR FLO, RTN MAINT FL	-	-	-	77	77	77		
4		RMT EXT RTC+ RMT EXT RTC- to REMOTE EXT O	-	-	-	49	49	49		
5		RM PWR MCLR+ RM PWR MCLR- to PWR MCLR REG	-	-	-	73	73	73		
6		MAINT PROC/W to MP I/F FAULT, RTN MP FAULT	-	-	-	175	175	175		
7		422 TR CK+WT, RTN TR CK+WT to RTN CHAN0 D, CHAN0 TRAN D	-	-	-	370	370	370		
8	tpd	423 TR+WT/RQ to CHAN1 TRAN D	-	-	-	370	370	370	nS	11a, 11b

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6.4 Module function. Maintenance processor serial interface. The maintenance processor serial interface (MPSI) module allows a computer, terminal or maintenance panel to take control of the AN/UYK-44 computer. The computer is controlled by operator manipulation of the maintenance processor module (BWAH). Interface of an external device or panel is via one of two independent full duplex channels. The MPSI contains two basic interfaces: the maintenance processor to MPSI interface and the MPSI to external device interface. The MPSI is capable of the following characteristics in synchronous mode:

(a) Up to one M bit per second data rate,
(b) Internal or external character synchronization,
(c) 1 or 2 sync characters
(d) Automatic sync character insertion and deletion
(e) 6- or 8-bit sync characters
(f) Transmit and receive clocks

The MPSI is capable of the following characteristics in asynchronous mode:

(a) 5-, 6-, 7-, or 8-bits per character
(b) 1,1-1/2 or 2 stop bits
(c) Odd or even parity
(d) 1, 16, 32 or 64 clocks per bit
(e) Parity, overrun and framing errors
(f) Data rates of 50, 75, 110, 134.5, 150, 300, 600, 1200, 2400, 3600, 4800, 7200, 9600 and 19,200 baud. (Default is 9600 baud).

The MPSI module contains the Universal Synchronous/Asynchronous Receive/Transmit (USART), USART memory, read/write registers, clock control logic, an input multiplexer, RX channel receivers, TRX channel receivers and drivers and TX channel drivers. The USART is a dual-channel serial communication controller which operates in both synchronous or asynchronous mode to control the MPSI. USART memory is composed of 4K of Random Access Memory (RAM) and 32K of Electrically Erasable Programmable Read Only Memory (EEPROM). Read and write registers are contained within the USART and are used to hold control, data and BIT information. TRX and RX receivers receive incoming data on channels 2/3 and 0/1, respectively. TRX and TX output drivers transmit data on channels 2/3 and 0/1/4, respectively. The input mux selects the USART data source. The clock and control logic is composed of a series of Programmable Logic Arrays (PALs) which control the USART clock input, decode address bits, control on-module

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registers and provide bus control. The module is under control of Remote Control Memory Interface (RCMI) firmware, which is contained in the USART memory.

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6.4.1 Description of I/O pins.

Signal name	Pin number	I/O	Description
MAP 1	11	I	MAP 1
USART CKSOUR	13	I	EXT USART PCLK SOURCE
MAINT PRSTAT	14	I	MAINT PROC STATUS (S1)
CPOSC FL IN	15	I	CP OSC FAULT
MAP 0	21	I	MAP 0
MAINT PR FLI	23	I	MAINT PROC FAULT
RMT INIT +	32	I	RMT INIT (NO MEM)
RMT INIT -	33	I	RMT INIT (NO MEM)
RMT EXT RTC +	34	I	REMOTE EXT RTC
RMT EXT RTC -	35	I	REMOTE EXT RTC
RMAUTO STRT +	36	I	REMOTE AUTO START
RMAUTO STRT -	37	I	REMOTE AUTO START
RM PWR MCLR +	38	I	RMT PWR MA CLR SIM
RM PWR MCLR -	39	I	RMT PWR MA CLR SIM
MP ADRS 15	42	I	MP ADDRESS 15
MP ADRS 14	43	I	MP ADDRESS 14
MP ADRS 13	44	I	MP ADDRESS 13
MP ADRS 12	45	I	MP ADDRESS 12
MP ADRS 11	46	I	MP ADDRESS 11
MP ADRS 10	47	I	MP ADDRESS 10
MP ADRS 9	48	I	MP ADDRESS 9

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MP ADRS 8	49	I	MP ADDRESS 8
MAINT PR ALE	61	I	MAINT PROC ALE
CHAN1 CK R/D	62	I	CHAN 1 CKT RD RCV DATA
RTN CHAN1 CK	63	I	(RTN) CHAN 1 CKT RD RCV DATA
CHAN1 RR RDY	64	I	CHAN 1 RR RCVR RDY
RTN CHAN1 RR	65	I	(RTN) CHAN 1 RR RCVR RDY
CHAN1 CLR SD	66	I	CHAN 1 CS CLR TO SEND
RTN CHAN1 SD	67	I	(RTN) CHAN 1 CS CLR TO SEND
CHAN1 D/MODE	68	I	CHAN 1 DM DATA MODE
RTN CHAN1 DM	69	I	(RTN) CHAN 1 DM DATA MODE
CHAN0 CK R/D	71	I	CHAN 0 CKT RD RCV DATA
RTN CHAN0 CK	72	I	(RTN) CHAN 0 CKT RD RCV DATA
CHAN0 RR RDY	73	I	CHAN 0 RR RCVR RDY
RTN CHAN0 RR	74	I	(RTN) CHAN 0 RR RCVR RDY
CHAN0 CLR SD	76	I	CHAN 0 CS CLR TO SEND
RTN CHAN0 SD	77	I	(RTN) CHAN 0 CS CLR TO SEND
CHAN0 D/MODE	78	I	CHAN 0 DM DATA MODE
RTN CHAN0 DM	79	I	(RTN) CHAN 0 DM DATA MODE
EPROM BYTE/W	80	I	EEPROM BYTE WRITE
PWR MA CLR	86	I	PWR MA CLR
OPTIONS EN 0	87	I	OPTIONS AVAIL ENBL 0

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
422 TR CKTXD	2	I/O	422 TX/RX CLK + TX DATA
RTN TR CKTXD	3	I/O	(RTN) TX/RX CLK + TXD
422 TR CK+WT	4	I/O	422 TX/RX CLK + WAIT
RTN TR CK+WT	5	I/O	(RTN) TX/RX CLK + WAIT
422 SYNC+D/R	6	I/O	422 SYNC + DTR/REQ
RTN SYNC+D/R	7	I/O	(RTN) SYNC + DTR/REQ
BITA DCT+RTS	8	I/O	BIT A DISCRETE + RTS
RTN BITA+RTS	9	I/O	(RTN) BIT A + RTS
MAINT PROC/W	12	I	MAINT PROC WRITE
423 TR CK+TX	16	I/O	423 TX/RX CLK + TX
423 TR+WT/RQ	17	I/O	423 TX/RX CLK + WAIT/REQ
423 SYNC+DR	18	I/O	423 SYNC + DTR/REQ
BITB DCT+RTS	19	I/O	BIT B DISCRETE + RTS
MAINT PROC/R	50	I	MAINT PROC READ
MP ADRS/D 7	52	I/O	MP ADRS/DATA 7
MP ADRS/D 6	53	I/O	MP ADRS/DATA 6
MP ADRS/D 5	54	I/O	MP ADRS/DATA 5
MP ADRS/D 4	55	I/O	MP ADRS/DATA 4
MP ADRS/D 3	56	I/O	MP ADRS/DATA 3
MP ADRS/D 2	57	I/O	MP ADRS/DATA 2
MP ADRS/D 1	58	I/O	MP ADRS/DATA 1

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MP ADRS/D 0	59	I/O	MP ADRS/DATA 0
REMOTE EXT O	22	O	REMOTE EXT RTC
RTN CHAN4 DO	24	O	(RTN) CHAN 4 CHAN A DATA OUT
CHAN4 D OUT	26	O	CHAN 4 DATA OUT
CPOSC FL OUT	27	O	CP OSC FAULT
RTN CPOSC FL	28	O	(RTN) CP OSC FAULT
MAINT PR FLO	29	O	MAINT PROC FAULT
RTN MAINT FL	30	O	(RTN) MAINT PROC FAULT
PWR MCLR REG	41	O	SIM PWR MA CLR TO RGTR
CHAN1 TRAN D	82	O	CHAN 1 TRANSMIT DATA
CHAN1 WT/REQ	83	O	CHAN 1 WAIT/REQ
CHAN1 DR/REQ	84	O	CHAN 1 DTR/REQ
CHAN1 REQ SD	85	O	CHAN 1 REQ TO SEND
MP I/F FAULT	88	O	MP SERIAL I/F FAULT
RTN MP FAULT	89	O	(RTN) MP SERIAL I/F FAULT
MP SI PR	91	O	MP SI PR (MI + MMIO 0 PR)
RTN CHAN0 D	92	O	(RTN) CHAN 0 TRANSMIT DATA
CHAN0 TRAN D	93	O	CHAN 0 TRANSMIT DATA
CHAN0 WT/REQ	94	O	CHAN 0 WAIT/REQ
RTN CHAN0 WT	95	O	(RTN) CHAN 0 WAIT/REQ
CHAN0 DR/REQ	96	O	CHAN 0 DTR/REQ

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
RTN CHAN0 DR	97	O	(RTN) CHAN 0 DTR/REQ
CHAN0 REQ SD	98	O	CHAN 0 REQ TO SEND
RTN CHAN0 SD	99	O	(RTN) CHAN 0 REQ TO SEND
USART INTRUP	100	O	USART INTERRUPT (RST, B)

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

*Custodians:
Army - ER
Navy - NW
Air Force - 85

*Review activities:
Army - AT
Navy - MC
Air Force - 13, 19
DLA - ES

*Preparing activity:
Navy - NW

(Project 5963-0868)

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TABLE II. Isolation test.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	C	Frame	10	INF	MOhm	1-8
A2	Isolation, frame	C	70	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1,A2
Isolation measurement	NA	± 500 kOhm	A1,A2

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

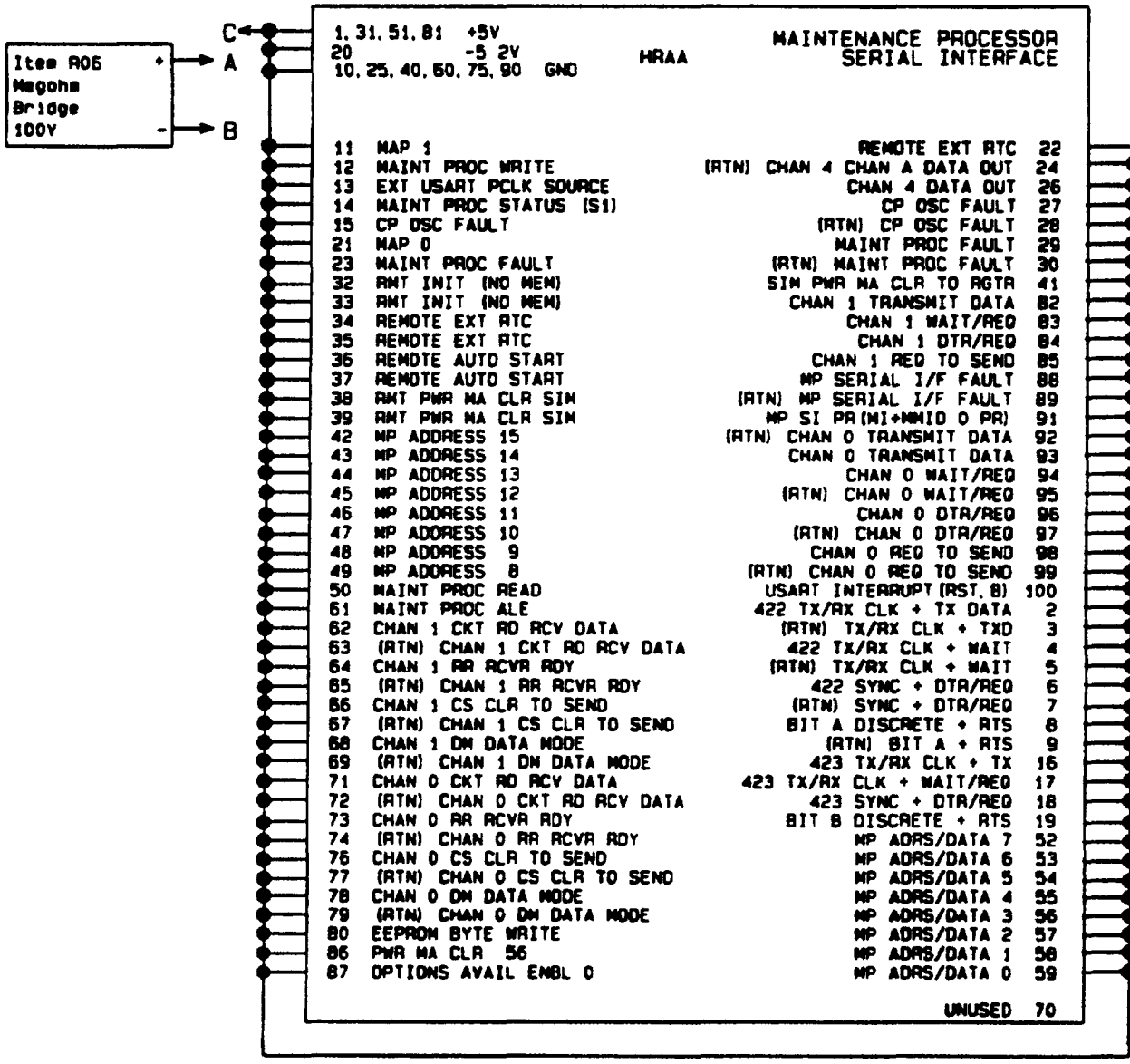
5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 μA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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* FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to pin	B to pin	Min	Max		
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	40	60				
B4	Continuity, GND	60	75				
B5	Continuity, GND	60	90				
B6	Continuity, Vcc	1	31				
B7	Continuity, Vcc	31	51				
B8	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B8

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1, 31, 51, 81 +5V
20 -5.2V
10, 25, 40, 60, 75, 90 GND

HRAA

MAINTENANCE PROCESSOR
SERIAL INTERFACE

11	MAP 1			REMOTE EXT RTC	22
12	MAINT PROC WRITE	(RTN)		CHAN 4 CHAN A DATA OUT	24
13	EXT USART PCLK SOURCE			CHAN 4 DATA OUT	26
14	MAINT PROC STATUS (S1)			CP OSC FAULT	27
15	CP OSC FAULT			(RTN) CP OSC FAULT	28
21	MAP 0			MAINT PROC FAULT	29
23	MAINT PROC FAULT			(RTN) MAINT PROC FAULT	30
32	RMT INIT (NO MEM)			SIM PWR MA CLR TO RGTR	41
33	RMT INIT (NO MEM)			CHAN 1 TRANSMIT DATA	82
34	REMOTE EXT RTC			CHAN 1 WAIT/REQ	83
35	REMOTE EXT RTC			CHAN 1 DTR/REQ	84
36	REMOTE AUTO START			CHAN 1 REQ TO SEND	85
37	REMOTE AUTO START			MP SERIAL I/F FAULT	88
38	RMT PWR MA CLR SIM			(RTN) MP SERIAL I/F FAULT	89
39	RMT PWR MA CLR SIM			MP SI PR(MI+MMIO 0 PR)	91
42	MP ADDRESS 15			(RTN) CHAN 0 TRANSMIT DATA	92
43	MP ADDRESS 14			CHAN 0 TRANSMIT DATA	93
44	MP ADDRESS 13			CHAN 0 WAIT/REQ	94
45	MP ADDRESS 12			(RTN) CHAN 0 WAIT/REQ	95
46	MP ADDRESS 11			CHAN 0 DTR/REQ	96
47	MP ADDRESS 10			(RTN) CHAN 0 DTR/REQ	97
48	MP ADDRESS 9			CHAN 0 REQ TO SEND	98
49	MP ADDRESS 8			(RTN) CHAN 0 REQ TO SEND	99
50	MAINT PROC READ			USART INTERRUPT (RST, B)	100
61	MAINT PROC ALE			422 TX/RX CLK + TX DATA	2
62	CHAN 1 CKT RD RCV DATA			(RTN) TX/RX CLK + TXD	3
63	(RTN) CHAN 1 CKT RD RCV DATA			422 TX/RX CLK + WAIT	4
64	CHAN 1 RR RCVR RDY			(RTN) TX/RX CLK + WAIT	5
65	(RTN) CHAN 1 RR RCVR RDY			422 SYNC + DTR/REQ	6
66	CHAN 1 CS CLR TO SEND			(RTN) SYNC + DTR/REQ	7
67	(RTN) CHAN 1 CS CLR TO SEND			BIT A DISCRETE + RTS	8
68	CHAN 1 DM DATA MODE			(RTN) BIT A + RTS	9
69	(RTN) CHAN 1 DM DATA MODE			423 TX/RX CLK + TX	16
71	CHAN 0 CKT RD RCV DATA			423 TX/RX CLK + WAIT/REQ	17
72	(RTN) CHAN 0 CKT RD RCV DATA			423 SYNC + DTR/REQ	18
73	CHAN 0 RR RCVR RDY			BIT B DISCRETE + RTS	19
74	(RTN) CHAN 0 RR RCVR RDY			MP ADRS/DATA 7	52
76	CHAN 0 CS CLR TO SEND			MP ADRS/DATA 6	53
77	(RTN) CHAN 0 CS CLR TO SEND			MP ADRS/DATA 5	54
78	CHAN 0 DM DATA MODE			MP ADRS/DATA 4	55
79	(RTN) CHAN 0 DM DATA MODE			MP ADRS/DATA 3	56
80	EEPROM BYTE WRITE			MP ADRS/DATA 2	57
86	PWR MA CLR 56			MP ADRS/DATA 1	58
87	OPTIONS AVAIL ENBL 0			MP ADRS/DATA 0	59
				UNUSED	70

* FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect		25 ± 5°C Initial limits	Notes
			Positive lead to	Negative lead to		
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6
C	V2	-7.0	20	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
(V2)	-7.0 V	-0 mV,+70 mV	C
Power supply current limit	2.94 A	NA	C
	0.40 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

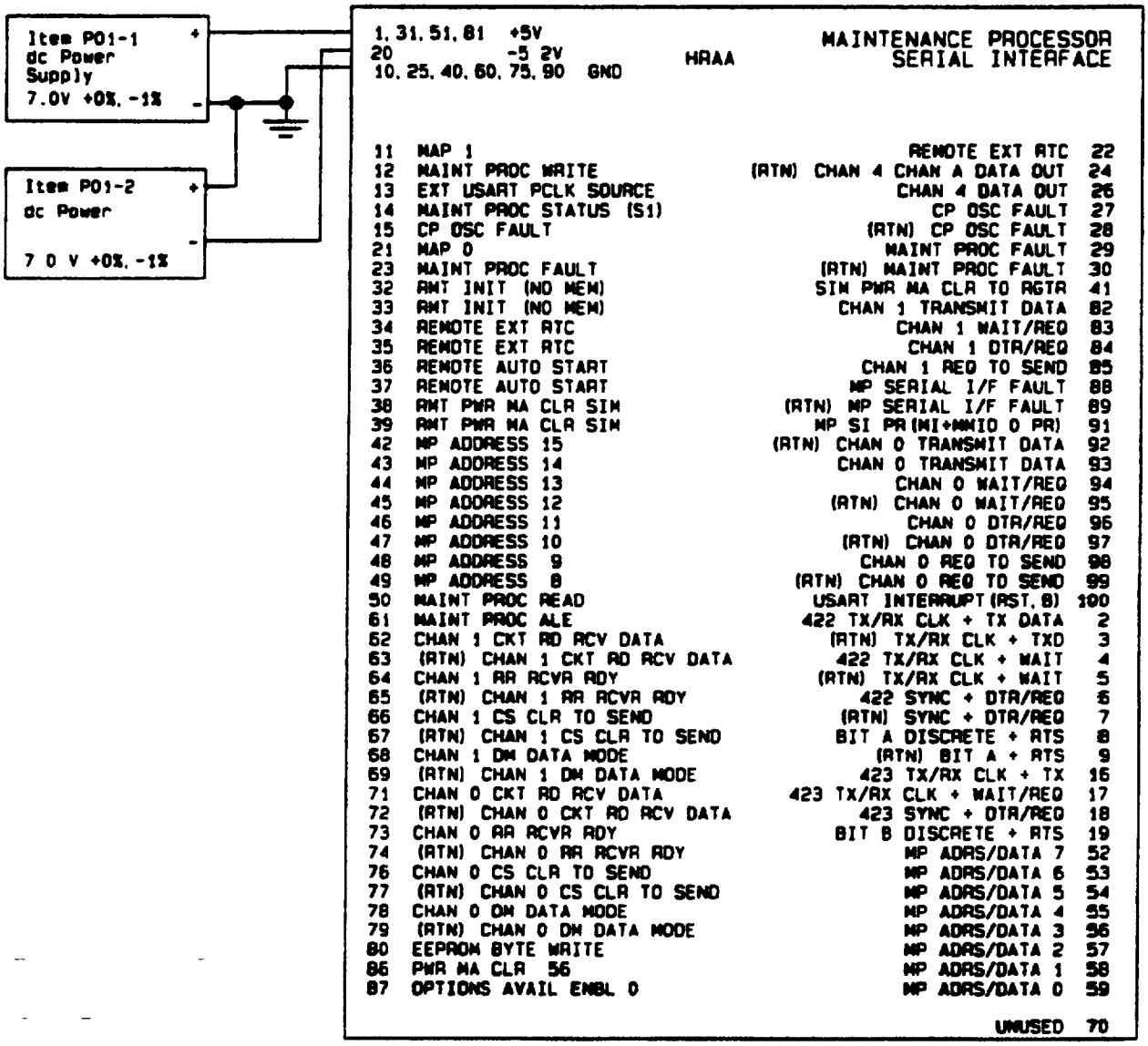
3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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* FIGURE 8. Vcc stress test circuit for table IV.

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* TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits		Units	Notes
				Min	Max		
D1	MAP 1	11	5.5	0.0	1.0	mA	1-4
D2	USART CKSOUR	13		-0.06	1.1		
D3	MAINT PRSTAT	14		0.0	0.1		
D4	CPOSC FL IN	15			0.1		
D5	MAP 0	21			1.0		
D6	MAINT PR FLI	23			0.1		
D7	MP ADRS 15	42					
D8	MP ADRS 14	43					
D9	MP ADRS 13	44					
D10	MP ADRS 12	45					
D11	MP ADRS 11	46					
D12	MP ADRS 10	47					
D13	MP ADRS 9	48					
D14	MP ADRS 8	49					
D15	MP ADRS/D 7	52					
D16	MP ADRS/D 6	53					
D17	MP ADRS/D 5	54					
D18	MP ADRS/D 4	55					
D19	MP ADRS/D 3	56					
D20	MP ADRS/D 2	57					
D21	MP ADRS/D 1	58	5.5	0.0	0.1	mA	1-4

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* TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits		Units	Notes
				Min	Max		
D22	MP ADRS/D 0	59	5.5	0.0	0.1	mA	1-4
D23	MAINT PR ALE	61		0.0	2.1		
D24	EPROM BYTE/W	80		-0.06	1.1		
D25	PWR MA CLR	86		0.0	2.3		
D26	OPTIONS EN 0	87	5.5	0.0	0.1	mA	1-4

NOTES:

*1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D26
(V2)	-5.72 V	± 57 mV	D1-D26
Power supply current limit (Vcc)	2.45 A	NA	D1-D26
(V2)	-0.033 A	NA	D1-D26
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D26
* Pattern voltage high, (VH)	See note 4	± 100 mV	D1-D26
* low, (VL)	See note 4	± 100 mV	D1-D26
Input current measurement	NA	± 50 µA	D1
		± 58 µA	D2
		± 5 µA	D3,D4
		± 50 µA	D5
		± 5 µA	D6-D22
		± 105 µA	D23
		± 58 µA	D24
		± 115 µA	D25
	NA	± 5 µA	D26

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

*4. The pattern voltage levels for pins 16-19 shall be +3.6 V for high level (VH) and -3.6 V for low level (VL). All other pins shall have pattern voltage levels of +3.0 V for high level (VH) and 0.0 V for low level (VL).

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* TABLE VI. Output short circuit current test.

	Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits		Units	Notes
					Min	Max		
	E1	422 TR CKTXD	2	0.0	-150.0	-6.0	mA	1-5
	E2	RTN TR CKTXD	3					
	E3	422 TR CK+WT	4					
	E4	RTN TR CK+WT	5					
	E5	422 SYNC+D/R	6					
	E6	RTN SYNC+D/R	7					
	E7	BITA DCT+RTS	8					
	E8	RTN BITA+RTS	9			-6.0		
	E9	423 TR CK+TX	16			-20.0		
	E10	423 TR+WT/RQ	17					
	E11	423 SYNC+DR	18					
	E12	BITB DCT+RTS	19		-150.0	-20.0		
	E13	REMOTE EXT O	22		-85.0	-15.0		
	E14	RTN CHAN4 DO	24		-76.4	-23.6		
	E15	CHAN4 D OUT	26					
	E16	CPOSC FL OUT	27					
	E17	RTN CPOSC FL	28					
	E18	MAINT PR FLO	29					
	E19	RTN MAINT FL	30		-76.4	-23.6		
*	E20	PWR MCLR REG	41		-2.92	-2.59		
*	E21	MP ADRS/D 7	52	0.0	-225.0	-40.0	mA	1-5

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* TABLE VI. Output short circuit current test - Continued.

	Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits		Units	Notes
					Min	Max		
*	E22	MP ADRS/D 6	53	0.0	-225.0	-40.0	mA	1-5
*	E23	MP ADRS/D 5	54					
*	E24	MP ADRS/D 4	55					
*	E25	MP ADRS/D 3	56					
*	E26	MP ADRS/D 2	57					
*	E27	MP ADRS/D 1	58					
*	E28	MP ADRS/D 0	59		-225.0	-40.0		
*	E29	CHAN1 TRAN D	82		-150.0	-20.0		
*	E30	CHAN1 WT/REQ	83					
*	E31	CHAN1 DR/REQ	84					
*	E32	CHAN1 REQ SD	85		-150.0	-20.0		
*	E33	MP I/F FAULT	88		-76.4	-23.6		
*	E34	RTN MP FAULT	89		-76.4	-23.6		
*	E35	RTN CHAN0 D	92		-150.0	-30.0		
*	E36	CHAN0 TRAN D	93					
*	E37	CHAN0 WT/REQ	94					
*	E38	RTN CHAN0 WT	95					
*	E39	CHAN0 DR/REQ	96					
*	E40	RTN CHAN0 DR	97					
*	E41	CHAN0 REQ SD	98					
*	E42	RTN CHAN0 SD	99	0.0	-150.0	-30.0	mA	1-5

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NOTES:

*1. Output short circuit current test tolerance table.

	Condition	Value	Accuracy	Test condition
*	Power supply, (Vcc)	5.5 V	± 55 mV	E1-E42
	(V2)	-5.72 v	± 57 mV	E1-E42
*	Power supply current (Vcc)	2.45 A	NA	E1-E42
	limit (V2)	-0.033 A	NA	E1-E42
*	Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E42
	Pattern voltage			
*	high, (VH)	See note 5	± 100 mV	E1-E42
*	low, (VL)	See note 5	± 100 mV	E1-E42
*	Output current measurement	NA	± 7.2 mA	E1-E8
			± 6.5 mA	E9-E12
			± 3.5 mA	E13
			± 2.6 mA	E14-E19
			± 0.017mA	E20
			± 9.3 mA	E21-E28
			± 6.5 mA	E29-E32
			± 2.6 mA	E33,E34
		NA	± 6.0 mA	E35-E42

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

*5. The pattern voltage levels for pins 16-19 shall be +3.6 V for high level (VH) and -3.6 V for low level (VL). All other pins shall have pattern voltage levels of +3.0 V for high level (VH) and 0.0 V for low level (VL).

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* TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits		Units	Notes
				Min	Max		
F1	422 TR CKTXD	2	2.4	-2.9	2.4	mA	1-4
F2	RTN TR CKTXD	3					
F3	422 TR CK+WT	4					
F4	RTN TR CK+WT	5					
F5	422 SYNC+D/R	6					
F6	RTN SYNC+D/R	7					
F7	BITA DCT+RTS	8					
F8	RTN BITA+RTS	9		-2.9	2.4		
F9	MAP 1	11		-0.25	0.025		
F10	MAINT PROC/W	12		-0.5	0.05		
F11	USART CKSOUR	13		-2.95	-1.41		
F12	MAINT PRSTAT	14		-0.1	0.02		
F13	CPOSC FL IN	15		-0.36	0.02		
F14	423 TR CK+TX	16		-2.9	2.4		
F15	423 TR+WT/RQ	17					
F16	423 SYNC+DR	18					
F17	BITB DCT+RTS	19		-2.9	2.4		
F18	MAP 0	21		-0.25	0.025		
F19	MAINT PR FLI	23		-0.36	0.02		
F20	RMT INIT +	32		-2.8	2.3		
F21	RMT INIT -	33		-2.8			
F22	RMT EXT RTC+	34	2.4	-2.8	2.3	mA	1-4

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* TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits		Units	Notes
				Min	Max		
F23	RMT EXT RTC-	35	2.4	-2.8	2.3	mA	1-4
F24	RMAUTO STRT+	36					
F25	RMAUTO STRT-	37					
F26	RM PWR MCLR+	38					
F27	RM PWR MCLR-	39		-2.8	2.3		
F28	MP ADRS 15	42		-0.2	0.02		
F29	MP ADRS 14	43					
F30	MP ADRS 13	44					
F31	MP ADRS 12	45					
F32	MP ADRS 11	46					
F33	MP ADRS 10	47					
F34	MP ADRS 9	48					
F35	MP ADRS 8	49		-0.2	0.02		
F36	MAINT PROC/R	50		-1.15	0.095		
F37	MP ADRS/D 7	52		-0.2	0.02		
F38	MP ADRS/D 6	53					
F39	MP ADRS/D 5	54					
F40	MP ADRS/D 4	55					
F41	MP ADRS/D 3	56					
F42	MP ADRS/D 2	57					
F43	MP ADRS/D 1	58	2.4	-0.2	0.02	mA	1-4

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* TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits		Units	Notes
				Min	Max		
F44	MP ADRS/D 0	59	2.4	-0.2	0.02	mA	1-4
F45	MAINT PR ALE	61		-0.7	0.07		
F46	CHAN1 CK R/D	62		-2.8	2.3		
F47	RTN CHAN1 CK	63					
F48	CHAN1 RR RDY	64					
F49	RTN CHAN1 RR	65					
F50	CHAN1 CLR SD	66					
F51	RTN CHAN1 SD	67					
F52	CHAN1 D/MODE	68					
F53	RTN CHAN1 DM	69					
F54	CHAN0 CK R/D	71					
F55	RTN CHAN0 CK	72					
F56	CHAN0 RR RDY	73					
F57	RTN CHAN0 RR	74					
F58	CHAN0 CLR SD	76					
F59	RTN CHAN0 SD	77					
F60	CHAN0 D/MODE	78					
F61	RTN CHAN0 DM	79		-2.8	2.3		
F62	EPROM BYTE/W	80		-2.7	-1.4		
F63	PWR MA CLR	86		-1.7	0.11		
F64	OPTIONS EN 0	87	2.4	-0.1	0.02	mA	1-4

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NOTES:

*1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F64
(V2)	-5.72 V	± 57 mV	F1-F64
Power supply current (Vcc) limit	2.45 A	NA	F1-F64
(V2)	-0.033 A	NA	F1-F64
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F64
* Pattern voltage high, (VH)	See note 4	± 100 mV	F1-F64
* low, (VL)	See note 4	± 100 mV	F1-F64
Input current measurement	NA	± 265 µA	F1-F8
		± 13.8 µA	F9
		± 27.5 µA	F10
		± 77 µA	F11
		± 6 µA	F12
		± 19 µA	F13
		± 265 µA	F14-F17
		± 13.8 µA	F18
		± 19 µA	F19
		± 255 µA	F20-F27
		± 11 µA	F28-F35
		± 62.3 µA	F36
		± 11 µA	F37-F44
		± 38.5 µA	F45
		± 255 µA	F46-F61
		± 65 µA	F62
		± 90.5 µA	F63
	NA	± 6 µA	F64

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

*4. The pattern voltage levels for pins 16-19 shall be +3.6 V for high level (VH) and -3.6 V for low level (VL). All other pins shall have pattern voltage levels of +3.0 V for high level (VH) and 0.0 V for low level (VL).

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* TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits		Units	Notes
				Min	Max		
G1	422 TR CKTXD	2	0.4	-2.9	2.4	mA	1-4
G2	RTN TR CKTXD	3					
G3	422 TR CK+WT	4					
G4	RTN TR CK+WT	5					
G5	422 SYNC+D/R	6					
G6	RTN SYNC+D/R	7					
G7	BITA DCT+RTS	8					
G8	RTN BITA+RTS	9		-2.9	2.4		
G9	MAP 1	11		-0.25	0.025		
G10	MAINT PROC/W	12		-0.5	0.05		
G11	USART CKSOUR	13		-2.95	-1.41		
G12	MAINT PRSTAT	14		-0.1	0.02		
G13	CPOSC FL IN	15		-0.36	0.02		
G14	423 TR CK+TX	16		-2.9	2.4		
G15	423 TR+WT/RQ	17					
G16	423 SYNC+DR	18					
G17	BITB DCT+RTS	19		-2.9	2.4		
G18	MAP 0	21		-0.25	0.025		
G19	MAINT PR FLI	23		-0.36	0.02		
G20	RMT INIT +	32		-2.8	2.3		
G21	RMT INIT -	33					
G22	RMT EXT RTC+	34	0.4	-2.8	2.3	mA	1-4

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* TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits		Units	Notes
				Min	Max		
G23	RMT EXT RTC-	35	0.4	-2.8	2.3	mA	1-4
G24	RMAUTO STRT+	36					
G25	RMAUTO STRT-	37					
G26	RM PWR MCLR+	38					
G27	RM PWR MCLR-	39		-2.8	2.3		
G28	MP ADRS 15	42		-0.2	0.02		
G29	MP ADRS 14	43					
G30	MP ADRS 13	44					
G31	MP ADRS 12	45					
G32	MP ADRS 11	46					
G33	MP ADRS 10	47					
G34	MP ADRS 9	48					
G35	MP ADRS 8	49		-0.2	0.02		
G36	MAINT PROC/R	50		-1.15	0.095		
G37	MP ADRS/D 7	52		-0.2	0.02		
G38	MP ADRS/D 6	53					
G39	MP ADRS/D 5	54					
G40	MP ADRS/D 4	55					
G41	MP ADRS/D 3	56					
G42	MP ADRS/D 2	57					
G43	MP ADRS/D 1	58	0.4	-0.2	0.02	mA	1-4

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* TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits		Units	Notes
				Min	Max		
G44	MP ADRS/D 0	59	0.4	-0.2	0.02	mA	1-4
G45	MAINT PR ALE	61		-0.7	0.07		
G46	CHAN1 CK R/D	62		-2.8	2.3		
G47	RTN CHAN1 CK	63					
G48	CHAN1 RR RDY	64					
G49	RTN CHAN1 RR	65					
G50	CHAN1 CLR SD	66					
G51	RTN CHAN1 SD	67					
G52	CHAN1 D/MODE	68					
G53	RTN CHAN1 DM	69					
G54	CHAN0 CK R/D	71					
G55	RTN CHAN0 CK	72					
G56	CHAN0 RR RDY	73					
G57	RTN CHAN0 RR	74					
G58	CHAN0 CLR SD	76					
G59	RTN CHAN0 SD	77					
G60	CHAN0 D/MODE	78					
G61	RTN CHAN0 DM	79		-2.8	2.3		
G62	EPROM BYTE/W	80		-2.7	-1.4		
G63	PWR MA CLR	86		-1.7	0.11		
G64	OPTIONS EN 0	87	0.4	-0.1	0.02	mA	1-4

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NOTES:

*1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G64
(V2)	-5.72 V	± 57 mV	G1-G64
Power supply current (Vcc) limit (V2)	2.45 A	NA	G1-G64
	-0.033 A	NA	G1-G64
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G64
* Pattern voltage high, (VH)	See note 4	± 100 mV	G1-G64
* low, (VL)	See note 4	± 100 mV	G1-G64
Input current measurement	NA	± 265 µA	G1-G8
		± 13.8 µA	G9
		± 27.5 µA	G10
		± 77 µA	G11
		± 6 µA	G12
		± 19 µA	G13
		± 265 µA	G14-G17
		± 13.8 µA	G18
		± 19 µA	G19
		± 255 µA	G20-G27
		± 11 µA	G28-G35
		± 62.3 µA	G36
		± 11 µA	G37-G44
		± 38.5 µA	G45
		± 255 µA	G46-G61
		± 65 µA	G62
		± 90.5 µA	G63
	NA	± 6 µA	G64

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

*4. The pattern voltage levels for pins 16-19 shall be +3.6 V for high level (VH) and -3.6 V for low level (VL). All other pins shall have pattern voltage levels of +3.0 V for high level (VH) and 0.0 V for low level (VL).

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* TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits		Units	Notes
				Min	Max		
H1	422 TR CKTXD	2	-0.16	2.0	6.0	V	1-5
H2	RTN TR CKTXD	3					
H3	422 TR CK+WT	4					
H4	RTN TR CK+WT	5					
H5	422 SYNC+D/R	6					
H6	RTN SYNC+D/R	7					
H7	BITA DCT+RTS	8					
H8	RTN BITA+RTS	9		2.0			
H9	423 TR CK+TX	16		3.6			
H10	423 TR+WT/RQ	17					
H11	423 SYNC+DR	18					
H12	BITB DCT+RTS	19		3.6	6.0		
H13	REMOTE EXT O	22	-0.16	2.4	4.5		
H14	RTN CHAN4 DO	24	-0.1				
H15	CHAN4 D OUT	26					
H16	CPOSC FL OUT	27					
H17	RTN CPOSC FL	28					
H18	MAINT PR FLO	29					
H19	RTN MAINT FL	30	-0.1				
H20	MP ADRS/D 7	52	-0.16	2.4	4.5	V	1-5

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* TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits		Units	Notes
				Min	Max		
H21	MP ADRS/D 6	53	-0.16	2.4	4.5	V	1-5
H22	MP ADRS/D 5	54					
H23	MP ADRS/D 4	55					
H24	MP ADRS/D 3	56					
H25	MP ADRS/D 2	57					
H26	MP ADRS/D 1	58					
H27	MP ADRS/D 0	59		2.4	4.5		
H28	CHAN1 TRAN D	82		3.6	6.0		
H29	CHAN1 WT/REQ	83					
H30	CHAN1 DR/REQ	84					
H31	CHAN1 REQ SD	85	-0.16	3.6	6.0		
H32	MP I/F FAULT	88	-0.1	2.4	4.5		
H33	RTN MP FAULT	89	-0.1				
H34	RTN CHAN0 D	92	-0.16				
H35	CHAN0 TRAN D	93					
H36	CHAN0 WT/REQ	94					
H37	RTN CHAN0 WT	95					
H38	CHAN0 DR/REQ	96					
H39	RTN CHAN0 DR	97					
H40	CHAN0 REQ SD	98					
H41	RTN CHAN0 SD	99	-0.16	2.4	4.5	V	1-5

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:17Z
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NOTES:

*1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H41
(V2)	-4.68 V	± 46 mV	H1-H41
Power supply current (Vcc)	2.45 A	NA	H1-H41
limit (V2)	-0.033 A	NA	H1-H41
Forcing current, (Ioh)	-0.16 mA	± 16 mA	H1-H13
	-0.1 mA	± 10 mA	H14-H19
	-0.16 mA	± 16 mA	H20-H31
	-0.1 mA	± 10 mA	H32,H33
	-0.16 mA	± 16 mA	H34-H41
* Pattern voltage high, (VH)	See note 5	± 100 mV	H1-H41
* low, (VL)	See note 5	± 100 mV	H1-H41
Output voltage measurement	NA	± 200 mV	H1-H8
		± 120 mV	H9-H12
		± 105 mV	H13-H27
		± 120 mV	H28-H31
	NA	± 105 mV	H32-H41

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

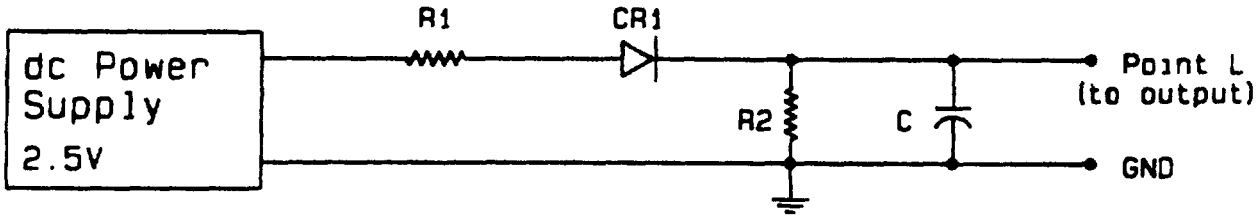
(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

*5. The pattern voltage levels for pins 16-19 shall be +3.6 V for high level (VH) and -3.6 V for low level (VL). All other pins shall have pattern voltage levels of +3.0 V for high level (VH) and 0.0 V for low level (VL).

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
12,22,50,52-59, 92-99	15,000	1.6	0.5
16-19,82-85	22,600	1.6	-6.0
24,26-30,88,89	24,300	1.0	0.5
41,91,100	NA	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

FIGURE 9. Logic load circuit.

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* TABLE X. Output voltage, low level, test.

	Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits		Units	Notes
					Min	Max		
*	J1	422 TR CKTXD	2	0.0	-6.0	-2.0	V	1-5
*	J2	RTN TR CKTXD	3					
*	J3	422 TR CK+WT	4					
*	J4	RTN TR CK+WT	5					
*	J5	422 SYNC+D/R	6					
*	J6	RTN SYNC+D/R	7					
*	J7	BITA DCT+RTS	8					
*	J8	RTN BITA+RTS	9	0.0		-2.0		
	J9	423 TR CK+TX	16	1.6		-3.6		
	J10	423 TR+WT/RQ	17					
	J11	423 SYNC+DR	18					
	J12	BITB DCT+RTS	19		-6.0	-3.6		
	J13	REMOTE EXT O	22	1.6	0.0	0.5		
	J14	RTN CHAN4 DO	24	1.0				
	J15	CHAN4 D OUT	26					
	J16	CPOSC FL OUT	27					
	J17	RTN CPOSC FL	28					
	J18	MAINT PR FLO	29					
	J19	RTN MAINT FL	30	1.0				
	J20	PWR MCLR REG	41	1.6				
	J21	MP ADRS/D 7	52					
	J22	MP ADRS/D 6	53	1.6	0.0	0.5	V	1-5

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* TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits		Units	Notes
				Min	Max		
J23	MP ADRS/D 5	54	1.6	0.0	0.5	V	1-5
J24	MP ADRS/D 4	55					
J25	MP ADRS/D 3	56					
J26	MP ADRS/D 2	57					
J27	MP ADRS/D 1	58					
J28	MP ADRS/D 0	59		0.0	0.5		
J29	CHAN1 TRAN D	82		-6.0	-3.6		
J30	CHAN1 WT/REQ	83					
J31	CHAN1 DR/REQ	84					
J32	CHAN1 REQ SD	85	1.6	-6.0	-3.6		
J33	MP I/F FAULT	88	1.0	0.0	0.5		
J34	RTN MP FAULT	89	1.0				
J35	MP SI PR	91	1.6				
J36	RTN CHAN0 D	92					
J37	CHAN0 TRAN D	93					
J38	CHAN0 WT/REQ	94					
J39	RTN CHAN0 WT	95					
J40	CHAN0 DR/REQ	96					
J41	RTN CHAN0 DR	97					
J42	CHAN0 REQ SD	98					
J43	RTN CHAN0 SD	99					
J44	USART INTRUP	100	1.6	0.0	0.5	V	1-5

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NOTES:

*1. Output voltage, low level, test tolerance table.

	Condition	Value	Accuracy	Test condition
	Power supply, (Vcc)	4.5 V	± 45 mV	J1-J44
	(V2)	-4.68 V	± 46 mV	J1-J44
	Power supply current (Vcc) limit	2.45 A	NA	J1-J44
	(V2)	-0.033 A	NA	J1-J44
*	Forcing current, (Iol)	0.0 mA	± 10 µA	J1-J8
		1.6 mA	± 160 µA	J9-J13
		1.0 mA	± 100 µA	J14-J19
		1.6 mA	± 160 µA	J20-J32
		1.0 mA	± 100 µA	J33,J34
		1.6 mA	± 160 µA	J35-J44
*	Pattern voltage high, (VH)	See note 5	± 100 mV	J1-J44
*	low, (VL)	See note 5	± 100 mV	J1-J44
	Output voltage measurement	NA	± 200 mV	J1-J8
			± 120 mV	J9-J12
			± 25 mV	J13-J28
			± 120 mV	J29-J32
		NA	± 25 mV	J33-J44

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

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NOTES: - Continued.

*3. The load shall be as indicated on figure 9. Pins 2-9 are outputs of differential drivers and shall have no load. The differential voltage is between pin pairs 2 and 3, 4 and 5, 6 and 7, and 8 and 9.

4. Abbreviations used are defined in 6.1.

*5. The pattern voltage levels for pins 16-19 shall be +3.6 V for high level (VH) and -3.6 V for low level (VL). All other pins shall have pattern voltage levels of +3.0 V for high level (VH) and 0.0 V for low level (VL).

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* TABLE XI. Output leakage current, high level, test.

	Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits		Units	Notes
					Min	Max		
*	K1	RTN CHAN4 DO	24	0.4	-0.02	0.02	mA	1-4
*	K2	CHAN4 D OUT	26					
*	K3	CPOSC FL OUT	27					
*	K4	RTN CPOSC FL	28					
*	K5	MAINT PR FLO	29					
*	K6	RTN MAINT FL	30					
*	K7	MP I/F FAULT	88					
*	K8	RTN MP FAULT	89					
*	K9	RTN CHAN0 D	92					
*	K10	CHAN0 TRAN D	93					
*	K11	CHAN0 WT/REQ	94					
*	K12	RTN CHAN0 WT	95					
*	K13	CHAN0 DR/REQ	96					
*	K14	RTN CHAN0 DR	97					
*	K15	CHAN0 REQ SD	98					
*	K16	RTN CHAN0 SD	99	0.4	-0.02	0.02		
*	K17	MP SI PR	91	5.5	0.0	0.01		
*	K18	USART INTRUP	100	5.5	0.0	0.01		
*	K19	RTN CHAN4 DO	24	2.4	-0.02	0.02		
*	K20	CHAN4 D OUT	26					
*	K21	CPOSC FL OUT	27	2.4	-0.02	0.02	mA	1-4

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*TABLE XI. Output leakage current, high level, test - Continued.

	Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits		Units	Notes
					Min	Max		
*	K22	RTN CPOSC FL	28	2.4	-0.02	0.02	mA	1-4
*	K23	MAINT PR FLO	29					
*	K24	RTN MAINT FL	30					
*	K25	MP I/F FAULT	88					
*	K26	RTN MP FAULT	89					
*	K27	RTN CHAN0 D	92					
*	K28	CHAN0 TRAN D	93					
*	K29	CHAN0 WT/REQ	94					
*	K30	RTN CHAN0 WT	95					
*	K31	CHAN0 DR/REQ	96					
*	K32	RTN CHAN0 DR	97					
*	K33	CHAN0 REQ SD	98					
*	K34	RTN CHAN0 SD	99	2.4	-0.02	0.02	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:17Z
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NOTES:

*1. Output leakage current, high level, test tolerance table.

	Condition	Value	Accuracy	Test condition
*	Power supply, (Vcc)	5.5 V	± 55 mV	K1-K34
	(V2)	-5.72 V	± 57 mV	K1-K34
*	Power supply current (Vcc)	2.45 V	NA	K1-K34
*	limit (V2)	-0.033 V	NA	K1-K34
*	Forcing voltage, (Vol)	0.4 V	± 4 mV	K1-K16
	(Voh)	5.5 V	± 55 mV	K17,K18
	(Voh)	2.4 V	± 24 mV	K19-K34
*	Pattern voltage high, (VH)	See note 4	± 100 mV	K1-K34
	low, (VL)	See note 4	± 100 mV	K1-K34
*	Output current	NA | NA	± 2.0 µA ± 0.5 µA ± 2.0 µA	K1-K16 K17,K18 K19-K34

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:17Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

*4. The pattern voltage levels for pins 16-19 shall be +3.6 V for high level (VH) and -3.6 V for low level (VL). All other pins shall have pattern voltage levels of +3.0 V for high level (VH) and 0.0 V for low level (VL).

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits		Units	Notes
				Min	Max		
L1	MAP 1	11	-18.0	-1.5	0.0	V	1-3
L2	MAINT PRSTAT	14					
L3	CPOSC FL IN	15					
L4	MAP 0	21					
L5	MAINT PR FLI	23					
L6	MP ADRS 15	42					
L7	MP ADRS 14	43					
L8	MP ADRS 13	44					
L9	MP ADRS 12	45					
L10	MP ADRS 11	46					
L11	MP ADRS 10	47					
L12	MP ADRS 9	48					
L13	MP ADRS 8	49					
L14	MP ADRS/D 7	52					
L15	MP ADRS/D 6	53					
L16	MP ADRS/D 5	54					
L17	MP ADRS/D 4	55					
L18	MP ADRS/D 3	56					
L19	MP ADRS/D 2	57					
L20	MP ADRS/D 1	58					
L21	MP ADRS/D 0	59					
L22	MAINT PR ALE	61	-18.0	-1.5	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits		Units	Notes
				Min	Max		
L23	PWR MA CLR	86	-18.0	-1.5	0.0	V	1-3
L24	OPTIONS EN 0	87	-18.0	-1.5	0.0	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2)	Off Off	NA NA	L1-L24 L1-L24
Power supply current limit	NA	NA	L1-L24
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L24
Pattern voltage high, (VH) low, (VL)	NA NA	NA NA	L1-L24 L1-L24
Input voltage measurement	NA	± 75 mV	L1-L24

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits		Units	Notes
			Min	Max		
M1	PSFC, Vcc	1	5.9	10.9	µF	1-4
M2	PSFC, V2	20	0.8	1.5	µF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

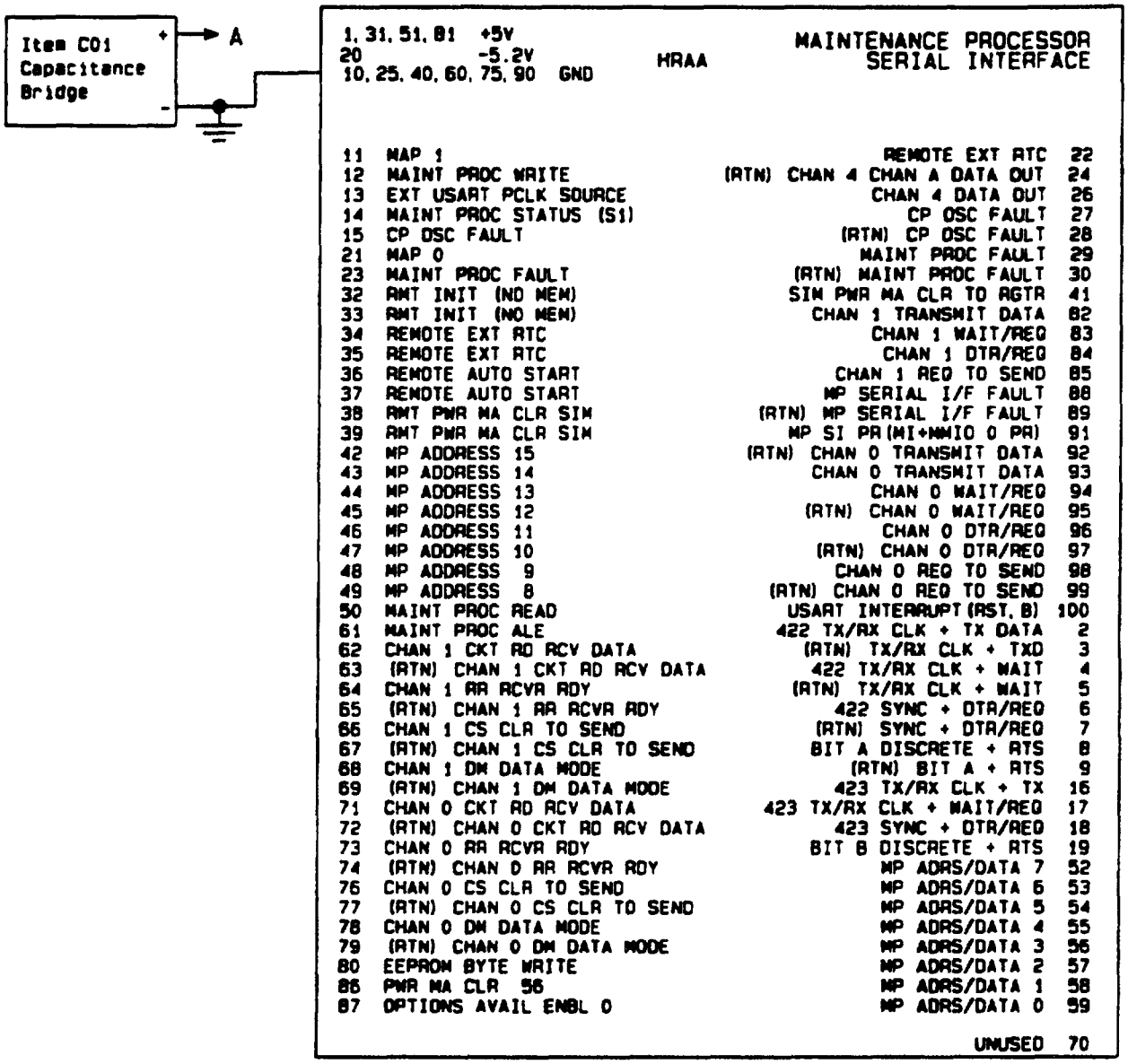
Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 250 nF ± 35 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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* FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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* TABLE XIV. Power supply current test.

	Test cond	Parameter	Pin number	25 ± 5°C Initial limits		Units	Notes
				Min	Max		
*	N1	Icc, Vcc	1	680	1633	mA	1-4
*	N2	Icc, V2	20	-22	-7	mA	1-4

NOTES:

*1. Power supply current test tolerance table.

	Condition	Value	Accuracy	Test condition
	Power supply, (Vcc)	5.5 V	± 55 mV	N1,N2
	(V2)	-5.72 V	± 57 mV	N1,N2
	Power supply current (Vcc)	2.45 A	NA	N1,N2
	limit (V2)	-0.033 A	NA	N1,N2
	Pattern voltage			
*	high, (VH)	See note 4	± 100 mV	N1,N2
*	low, (VL)	See note 4	± 100 mV	N1,N2
	Power supply current measurement	NA	± 47.6 mA	N1
		NA	± 0.75 mA	N2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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NOTES: - Continued.

*4. The pattern voltage levels for pins 16-19 shall be +3.6 V for high level (VH) and -3.6 V for low level (VL). All other pins shall have pattern voltage levels of +3.0 V for high level (VH) and 0.0 V for low level (VL).

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* TABLE XV. Functional test.

	Test cond	Test description	Notes
*	P1	Functional test minimum power supply voltage maximum pattern rate (3750 nS)	1-6
*	P2	Functional test maximum power supply voltage maximum pattern rate (3750 nS)	1-6

NOTES:

*1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	P1
-	5.5 V	± 55 mV	P2
(V2)	-4.68 V	± 46 mV	P1
	-5.72 V	± 57 mV	P2
Power supply current (Vcc)	2.45 A	NA	P1,P2
limit (V2)	-0.033 A	NA	P1,P2
Pattern voltage			
* high, (VH)	See note 4	± 100 mV	P1,P2
* low, (VL)	See note 4	± 100 mV	P1,P2
Output voltage verification			
high, (Voh)	NA	± 105 mV	P1
high, (Voh)	|	± 155 mV	P2
low, (Vol)	NA	± 20 mV	P1,P2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

* (b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

Pins 16-19, 82-85:
High level: 3.56 V $\leq$ Voh $\leq$ 6.0 V
Low level: -6.0 V $\leq$ Vol $\leq$ -3.56 V

All other pins
High level: 2.4 V $\leq$ Voh $\leq$ Vcc
Low level: 0.0 V $\leq$ Vol $\leq$ 0.5 V

3. Abbreviations used are defined in 6.1.

*4. The pattern voltage levels for pins 16-19 shall be +3.6 V for high level (VH) and -3.6 V for low level (VL). All other pins shall have pattern voltage levels of +2.4 V for high level (VH) and 0.4 V for low level (VL).

*5. The first part of the oscillator test verifies the input/output path by applying an external 1.0 MHz, 4.0 V (0.0 V dc offset) signal at input pin 13 and measuring a signal with a frequency between 75.00 kHz and 91.67 kHz at output pins 2, 3, 4, 5, 16 and 17. The input signal is a positive going square waveform with a 50 percent duty cycle.

The second part of the test verifies the internal oscillator operation and the same path by measuring a signal with a frequency between 276.450 kHz and 337.883 kHz at output pins 2, 3, 4, 5, 16 and 17.

*6. The static functional test shall consist of a test to ensure that, for each memory cell, no address faults, stuck-at-faults or data faults exist.

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* TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (3750 nS) minimum power supply	1-7
Q2	Dynamic test, sample maximum pattern rate (3750 nS) maximum power supply	1-7
Q3	Dynamic test, qualification maximum pattern rate (3750 nS) minimum power supply	1-7
Q4	Dynamic test, qualification maximum pattern rate (3750 nS) maximum power supply	1-7

NOTES:

*1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	Q1,Q3
	5.5 V	± 55 mV	Q2,Q4
(V2)	-4.68 V	± 46 mV	Q1,Q3
	-5.72 V	± 57 mV	Q2,Q4
Power supply current (Vcc)	2.45 A	NA	Q1-Q4
limit (V2)	-0.033 A	NA	Q1-Q4
* Input voltage high, (VH)	See note 7	± 100 mV	Q1-Q4
* low, (VL)	See note 7	± 100 mV	Q1-Q4
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	|	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

* (b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

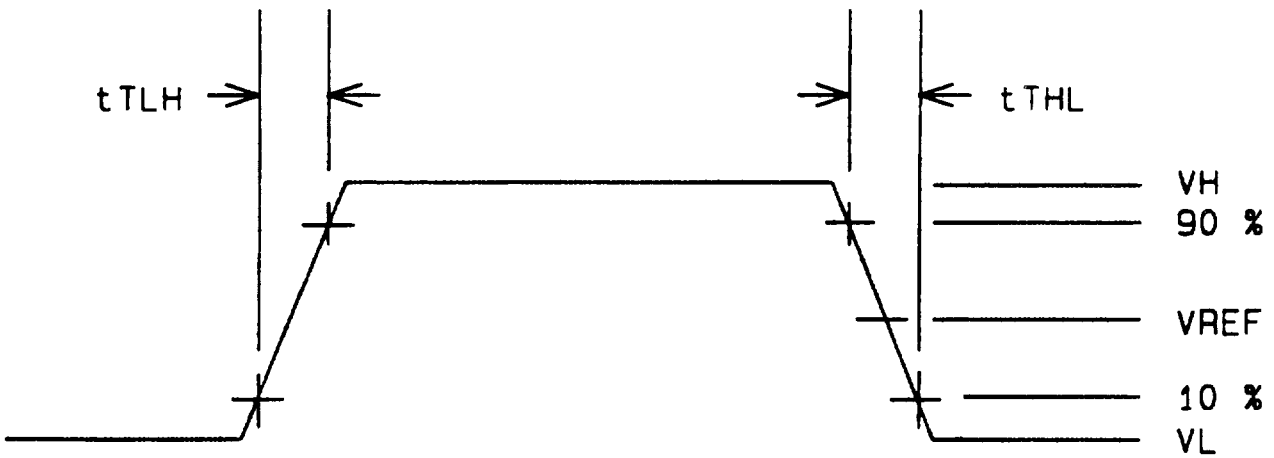
Pins 16-19, 82-85:
High level: Vref $\leq$ Voh $\leq$ Vcc
Low level: V2 $\leq$ Vol $\leq$ Vref

All other pins:
High level: 2.4 V $\leq$ Voh $\leq$ Vcc
Low level: 0.0 V $\leq$ Vol $\leq$ 0.5 V

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 11. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

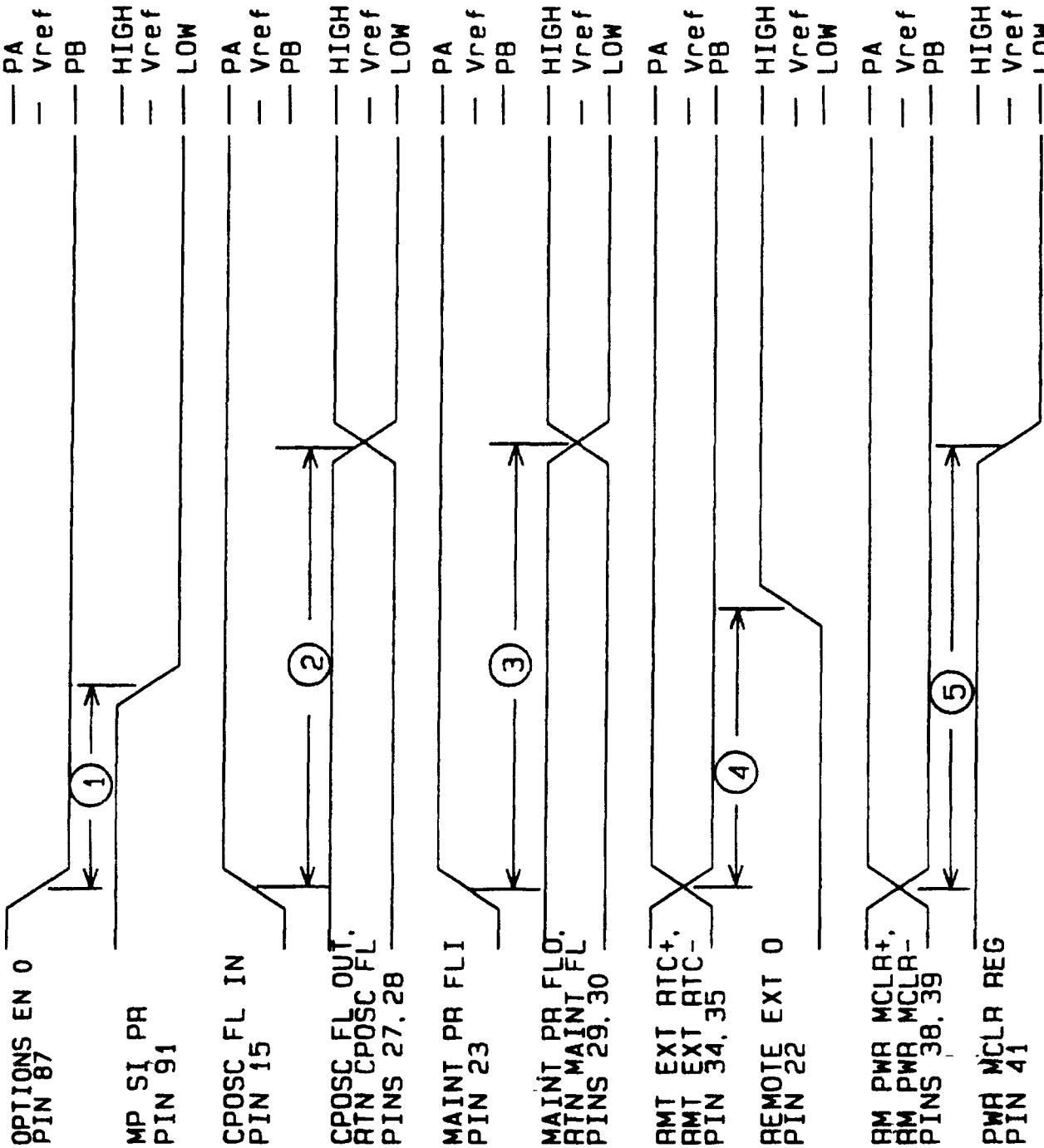
*7. The input voltage levels for pins 16-19 shall be +3.6 V for high level (VH) and -3.6 V for low level (VL). All other pins shall have pattern voltage levels of +2.4 V for high level (VH) and 0.4 V for low level (VL).

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TABLE XVII. Dynamic parameters.

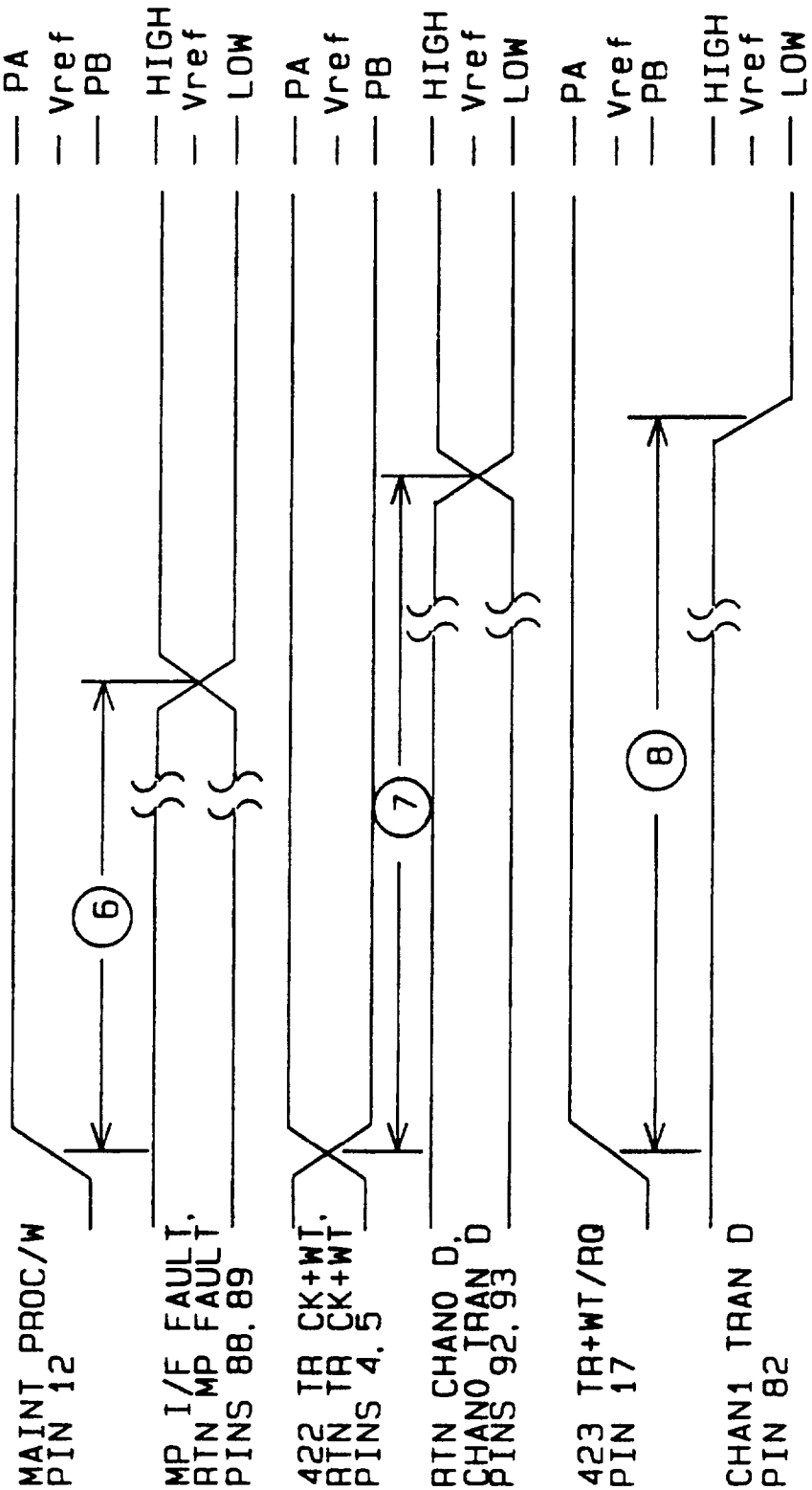
Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	OPTIONS EN 0 to MP SI PR	-	-	-	31	31	31	nS	11a, 11b
2		CPOSC FL IN to CPOSC FL OUT, RTN MAINT FL	-	-	-	77	77	77		
3		MAINT PR FLI to MAINT PR FLO, RTN MAINT FL	-	-	-	77	77	77		
4		RMT EXT RTC+, RMT EXT RTC- to REMOTE EXT O	-	-	-	49	49	49		
5		RM PWR MCLR+ RM PWR MCLR- to PWR MCLR REG	-	-	-	73	73	73		
6		MAINT PROC/W to MP I/F FAULT, RTN MP FAULT	-	-	-	175	175	175		
7		422 TR CK+WT, RTN TR CK+WT to RTN CHAN0 D, CHAN0 TRAN D	-	-	-	370	370	370		
8	tpd	423 TR+WT/RQ to CHAN1 TRAN D	-	-	-	370	370	370	nS	11a, 11b

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* FIGURE 11a. Dynamic waveforms for table XVII.

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* **FIGURE 11b.** Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits		Units	Notes
			Min	Max		
R1	MAP 1	11	2	50	pF	1-4
R2	USART CKSOUR	13	4	75		
R3	MAINT PRSTAT	14	2			
R4	CPOSC FL IN	15				
R5	MAP 0	21				
R6	MAINT PR FLI	23		75		
R7	RMT INIT +	32		50		
R8	RMT INIT -	33				
R9	RMT EXT RTC +	34				
R10	RMT EXT RTC -	35				
R11	RMAUTO STRT +	36				
R12	RMAUTO STRT -	37				
R13	RM PWR MCLR +	38				
R14	RM PWR MCLR -	39				
R15	MP ADRS 15	42				
R16	MP ADRS 14	43				
R17	MP ADRS 13	44				
R18	MP ADRS 12	45				
R19	MP ADRS 11	46				
R20	MP ADRS 10	47				
R21	MP ADRS 9	48	2	50	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits		Units	Notes
			Min	Max		
R22	MP ADRS 8	49	2	50	pF	1-4
R23	MP ADRS/D 7	52				
R24	MP ADRS/D 6	53				
R25	MP ADRS/D 5	54				
R26	MP ADRS/D 4	55				
R27	MP ADRS/D 3	56		50		
R28	MP ADRS/D 2	57		75		
R29	MP ADRS/D 1	58		50		
R30	MP ADRS/D 0	59		50		
R31	MAINT PR ALE	61		125		
R32	CHAN1 CK R/D	62		50		
R33	RTN CHAN1 CK	63				
R34	CHAN1 RR RDY	64				
R35	RTN CHAN1 RR	65				
R36	CHAN1 CLR SD	66				
R37	RTN CHAN1 SD	67				
R38	CHAN1 D/MODE	68				
R39	RTN CHAN1 DM	69				
R40	CHAN0 CK R/D	71				
R41	RTN CHAN0 CK	72				
R42	CHAN0 RR RDY	73	2	50	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits		Units	Notes
			Min	Max		
R43	RTN CHAN0 RR	74	2	50	pF	1-4
R44	CHAN0 CLR SD	76				
R45	RTN CHAN0 SD	77				
R46	CHAN0 D/MODE	78				
R47	RTN CHAN0 DM	79	2	50		
R48	EPROM BYTE/W	80	4	200		
R49	PWR MA CLR	86	8	200		
R50	OPTIONS EN 0	87	2	75		
R51	422 TR CKTXD	2	4			
R52	RTN TR CKTXD	3				
R53	422 TR CK+WT	4				
R54	RTN TR CK+WT	5				
R55	422 SYNC+D/R	6				
R56	RTN SYNC+D/R	7				
R57	BITA DCT+RTS	8				
R58	RTN BITA+RTS	9				
R59	MAINT PROC/W	12	4	75		
R60	423 TR CK+TX	16	6	100		
R61	423 TR+WT/RQ	17				
R62	423 SYNC+DR	18				
R63	BITB DCT+RTS	19	6	100	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits		Units	Notes
			Min	Max		
R64	MAINT PROC/R	50	8	175	pF	1-4
R65	CHAN1 TRAN D	82	4	75		
R66	CHAN1 WT/REQ	83				
R67	CHAN1 DR/REQ	84				
R68	CHAN1 REQ SD	85	4	75	pF	1-4

NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R68
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R68
Capacitance measurement	NA	± 2.4 pF	R1
		± 3.55 pF	R2
		± 3.65 pF	R3-R6
		± 2.4 pF	R7-R27
		± 3.65 pF	R28
		± 2.4 pF	R29,R30
		± 6.15 pF	R31
		± 2.4 pF	R32-R47
		± 9.8 pF	R48
		± 9.6 pF	R49
		± 3.65 pF	R50
		± 3.55 pF	R51-R59
		± 4.7 pF	R60-R63
		± 8.35 pF	R64
	NA	± 3.55 pF	R65-R68

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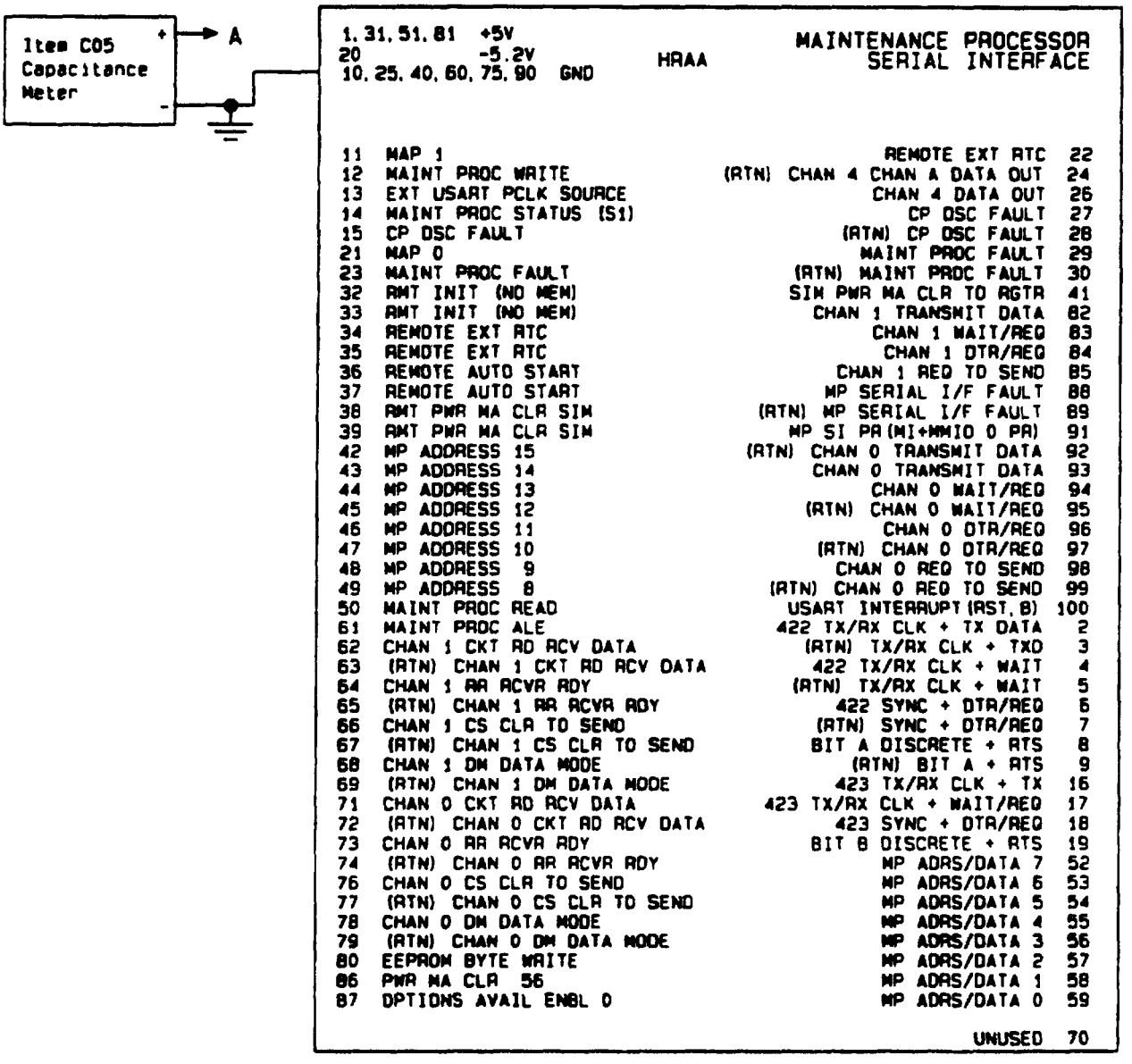
NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 12.

4. Abbreviations used are defined in 6.1.

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* FIGURE 12. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits		Units	Notes
			Min	Max		
T1	PSTA, Vcc	1	0	400	mV	1-4
T2	PSTA, V2	20	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2)	5.0 V -5.2 V	± 250 mV ± 52 mV	T1 T2
Power supply current (Vcc) limit (V2)	2.45 A -0.033 A	NA NA	T1,T2 T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

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NOTES: - Continued.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 13 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 14.

4. Abbreviations used are defined in 6.1.

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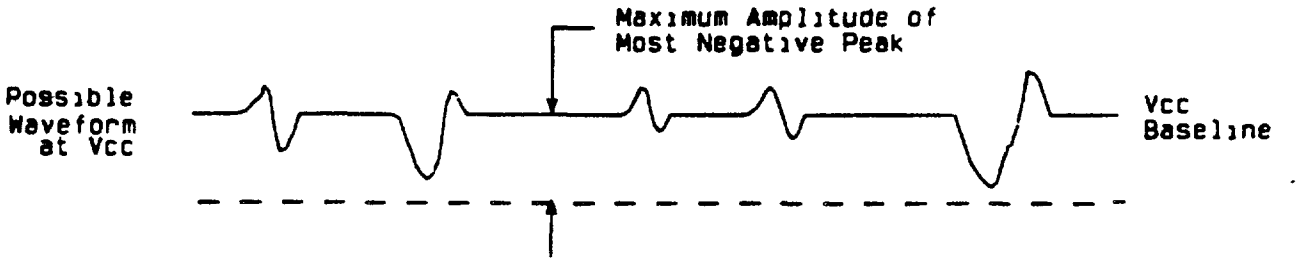
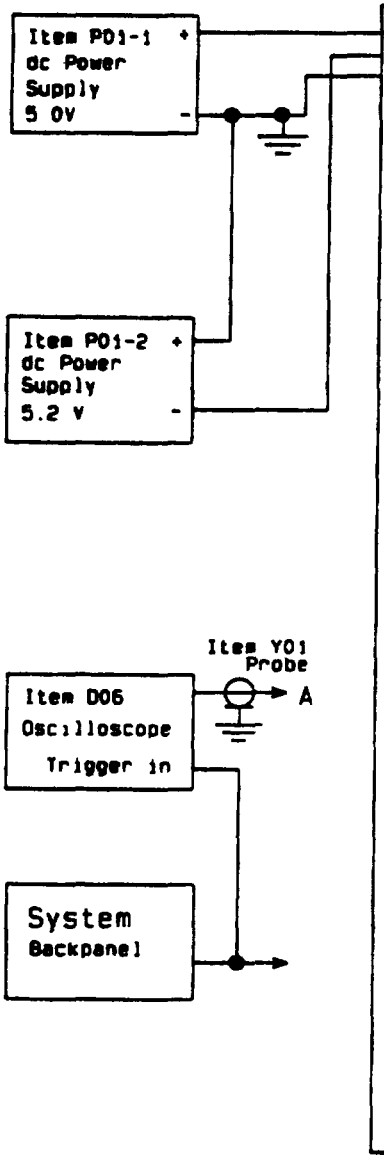


FIGURE 13. Power supply transient amplitude waveform for table XIX.

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1, 31, 51, 81 +5V
20 -5.2V
10, 25, 40, 60, 75, 90 GND

HRAA

MAINTENANCE PROCESSOR
SERIAL INTERFACE

11	MAP 1			REMOTE EXT RTC	22
12	MAINT PROC WRITE		(RTN)	CHAN 4 CHAN A DATA OUT	24
13	EXT USART PCLK SOURCE			CHAN 4 DATA OUT	26
14	MAINT PROC STATUS (S1)			CP OSC FAULT	27
15	CP OSC FAULT		(RTN)	CP OSC FAULT	28
21	MAP 0			MAINT PROC FAULT	29
23	MAINT PROC FAULT		(RTN)	MAINT PROC FAULT	30
32	RMT INIT (NO MEM)			SIM PWR MA CLR TO RGTR	41
33	RMT INIT (NO MEM)			CHAN 1 TRANSMIT DATA	82
34	REMOTE EXT RTC			CHAN 1 WAIT/REQ	83
35	REMOTE EXT RTC			CHAN 1 DTR/REQ	84
36	REMOTE AUTO START			CHAN 1 REQ TO SEND	85
37	REMOTE AUTO START			MP SERIAL I/F FAULT	88
38	RMT PWR MA CLR SIM		(RTN)	MP SERIAL I/F FAULT	89
39	RMT PWR MA CLR SIM			MP SI PR(MI+MMIO 0 PR)	91
42	MP ADDRESS 15		(RTN)	CHAN 0 TRANSMIT DATA	92
43	MP ADDRESS 14			CHAN 0 TRANSMIT DATA	93
44	MP ADDRESS 13			CHAN 0 WAIT/REQ	94
45	MP ADDRESS 12		(RTN)	CHAN 0 WAIT/REQ	95
46	MP ADDRESS 11			CHAN 0 DTR/REQ	96
47	MP ADDRESS 10		(RTN)	CHAN 0 DTR/REQ	97
48	MP ADDRESS 9			CHAN 0 REQ TO SEND	98
49	MP ADDRESS 8		(RTN)	CHAN 0 REQ TO SEND	99
50	MAINT PROC READ			USART INTERRUPT (RST, 8)	100
61	MAINT PROC ALE			422 TX/RX CLK + TX DATA	2
62	CHAN 1 CKT RD RCV DATA		(RTN)	TX/RX CLK + TXD	3
63	(RTN) CHAN 1 CKT RD RCV DATA			422 TX/RX CLK + WAIT	4
64	CHAN 1 RR RCVR RDY		(RTN)	TX/RX CLK + WAIT	5
65	(RTN) CHAN 1 RR RCVR RDY			422 SYNC + DTR/REQ	6
66	CHAN 1 CS CLR TO SEND		(RTN)	SYNC + DTR/REQ	7
67	(RTN) CHAN 1 CS CLR TO SEND			BIT A DISCRETE + RTS	8
68	CHAN 1 DM DATA MODE		(RTN)	BIT A + RTS	9
69	(RTN) CHAN 1 DM DATA MODE			423 TX/RX CLK + TX	16
71	CHAN 0 CKT RD RCV DATA		423	TX/RX CLK + WAIT/REQ	17
72	(RTN) CHAN 0 CKT RD RCV DATA			423 SYNC + DTR/REQ	18
73	CHAN 0 RR RCVR RDY			BIT B DISCRETE + RTS	19
74	(RTN) CHAN 0 RR RCVR RDY			MP ADRS/DATA 7	52
76	CHAN 0 CS CLR TO SEND			MP ADRS/DATA 6	53
77	(RTN) CHAN 0 CS CLR TO SEND			MP ADRS/DATA 5	54
78	CHAN 0 DM DATA MODE			MP ADRS/DATA 4	55
79	(RTN) CHAN 0 DM DATA MODE			MP ADRS/DATA 3	56
80	EEPROM BYTE WRITE			MP ADRS/DATA 2	57
86	PWR MA CLR 56			MP ADRS/DATA 1	58
87	OPTIONS AVAIL ENBL 0			MP ADRS/DATA 0	59
				UNUSED	70

NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

* FIGURE 14. Power supply transient amplitude test circuit for table XIX.

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a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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8. PREPARING ACTIVITY

a. NAME Technical Point of Contact (TPOC): Mr. Joseph P. Toy (Code 603A) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial TPOC: 812-854-5993 (2) AUTOVON 88-482-5993
c. ADDRESS (Include Zip Code) Commander ATTN: Joseph P. Toy, Code 603A, Bldg. 2917 NAVSURFWARCENDIV, 300 Highway 361 Crane, IN 47522-5001	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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